Containers & OpenShift Workshop

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containers are created from container images

Let’s do a recap of kubernetes

1. RECAP

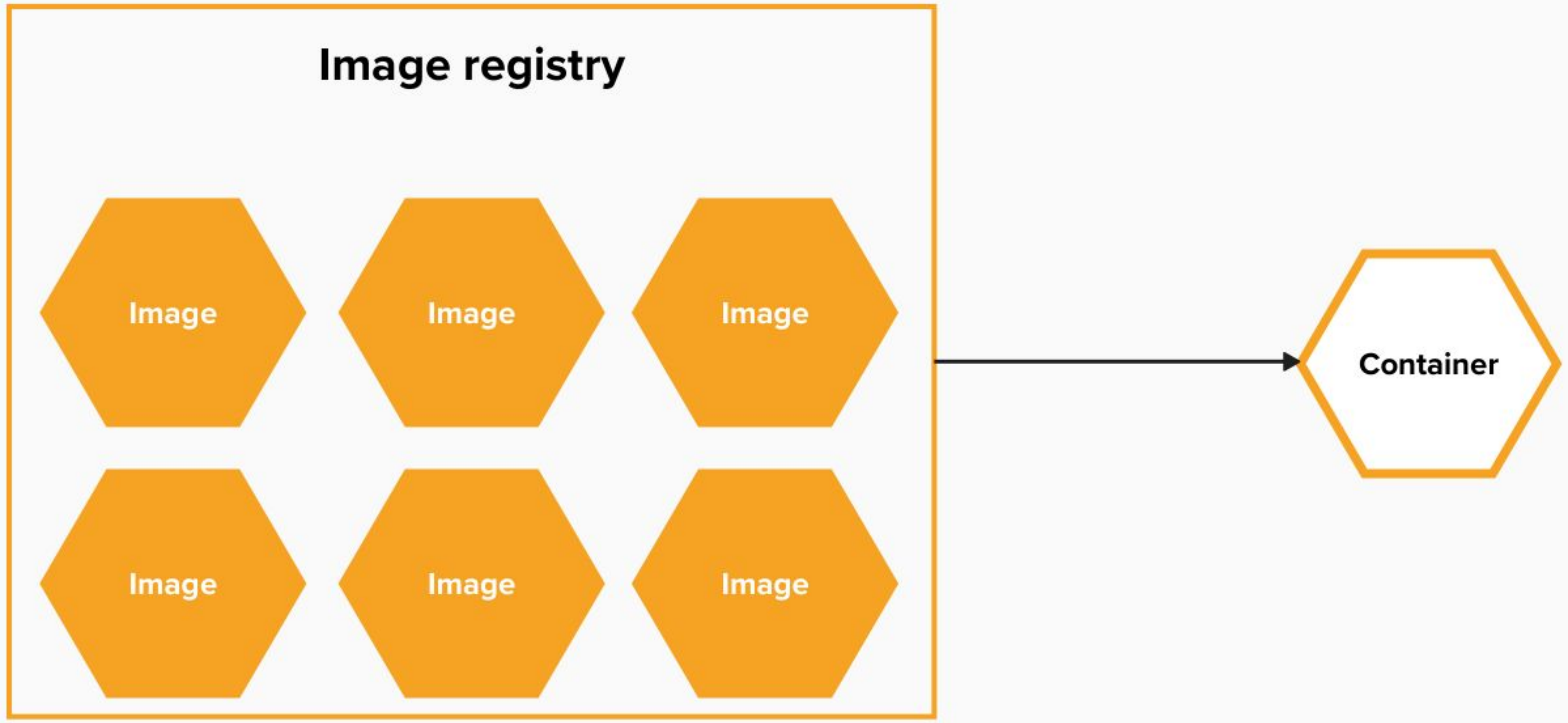
a container is the smallest compute unit

1. RECAP

containers are created from container images

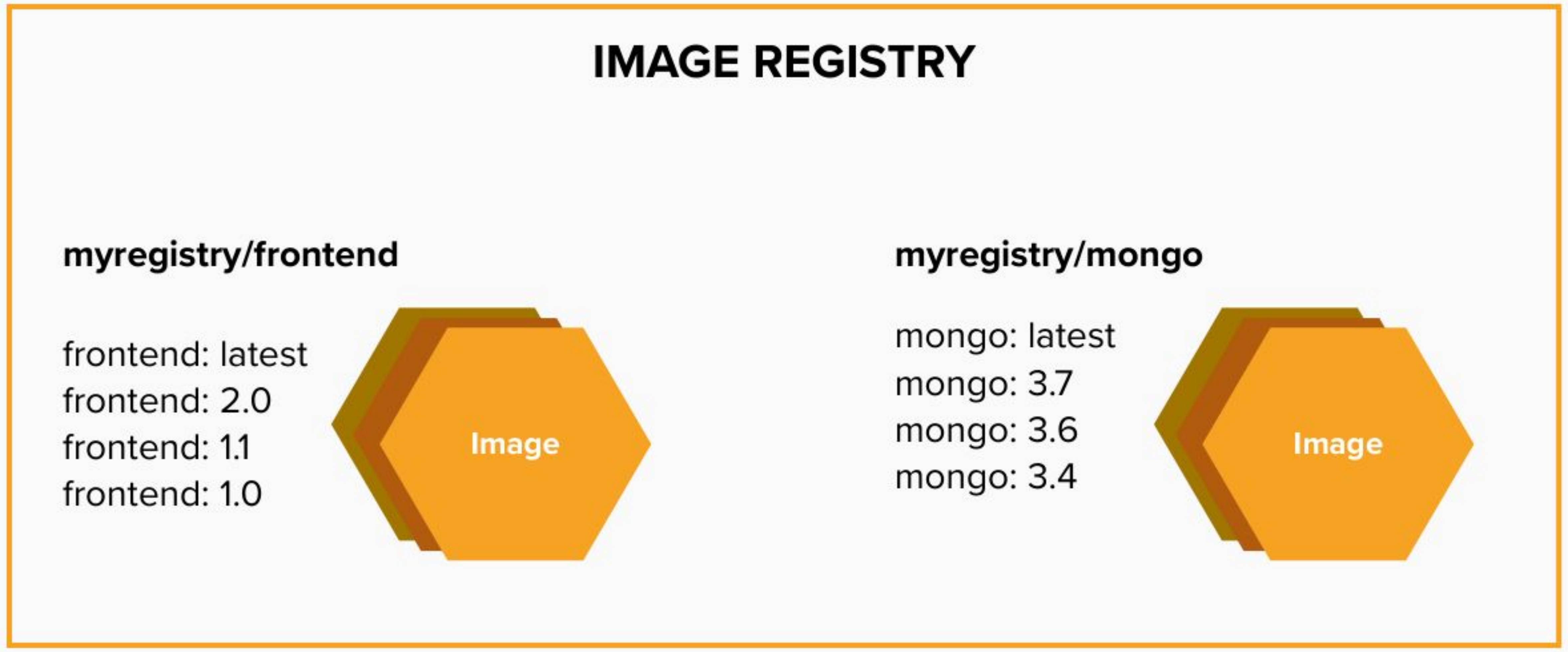
1. RECAP

container images are stored in an image registry



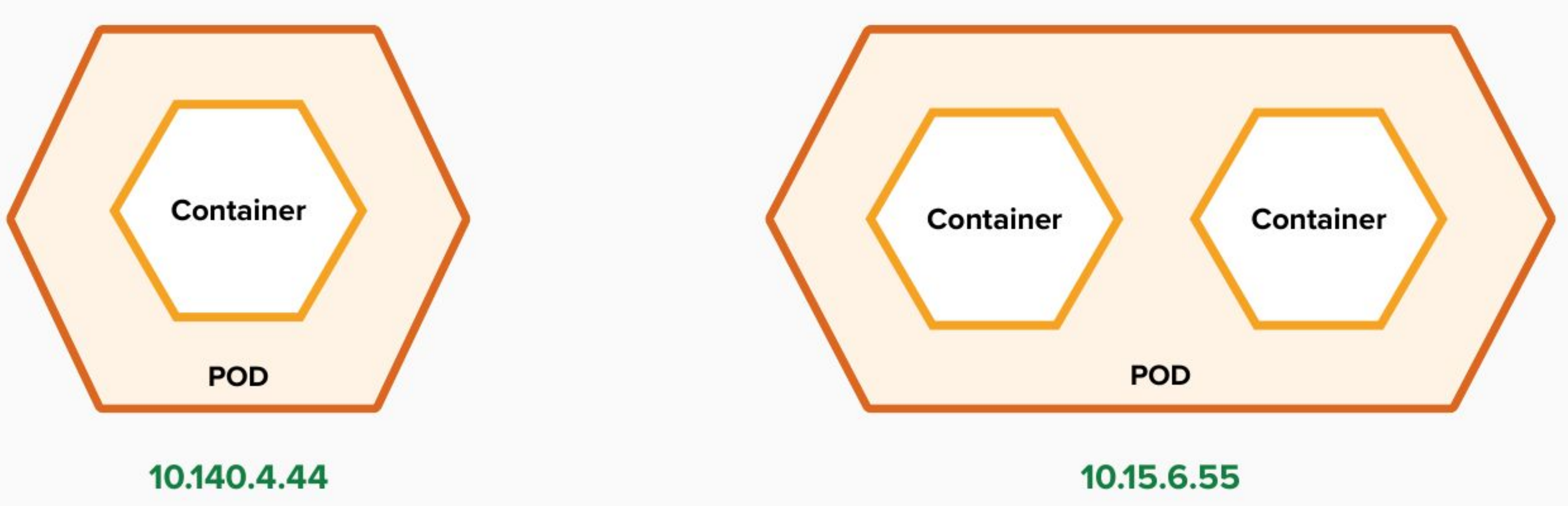
1. RECAP

an image repository contains all versions of an image in the image registry



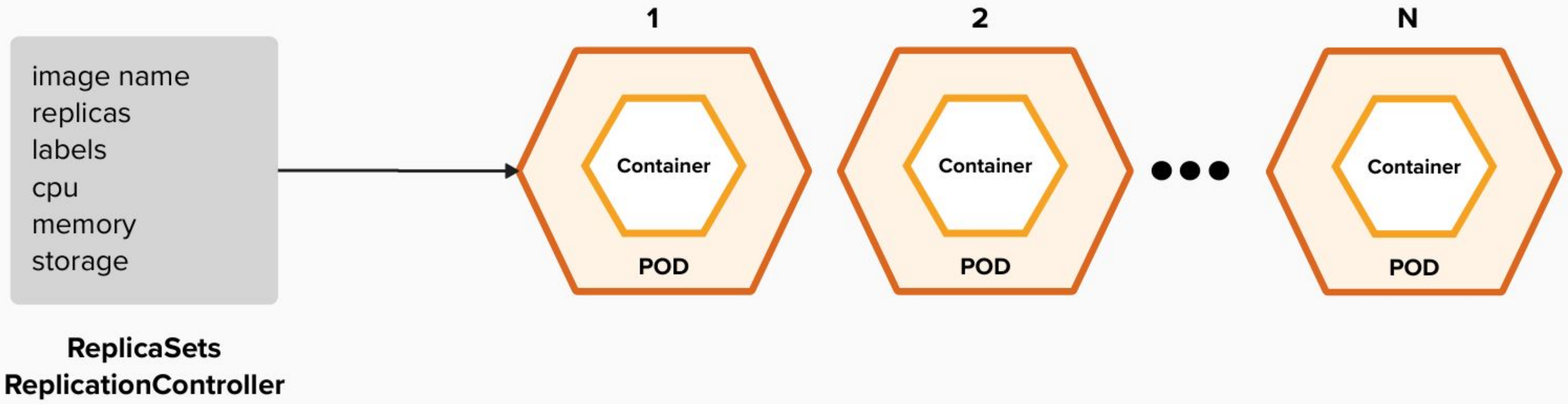
1. RECAP

containers are wrapped in pods which are units of deployment and management



1. RECAP

ReplicationControllers & ReplicaSets ensure a specified number of pods are running at any given time



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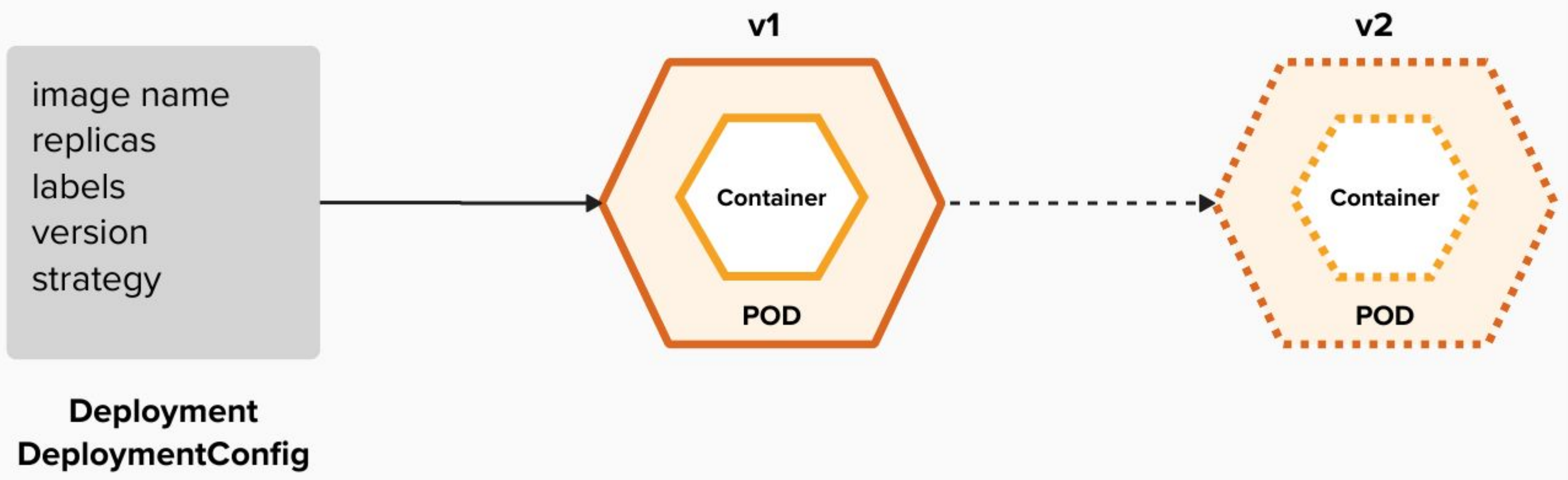
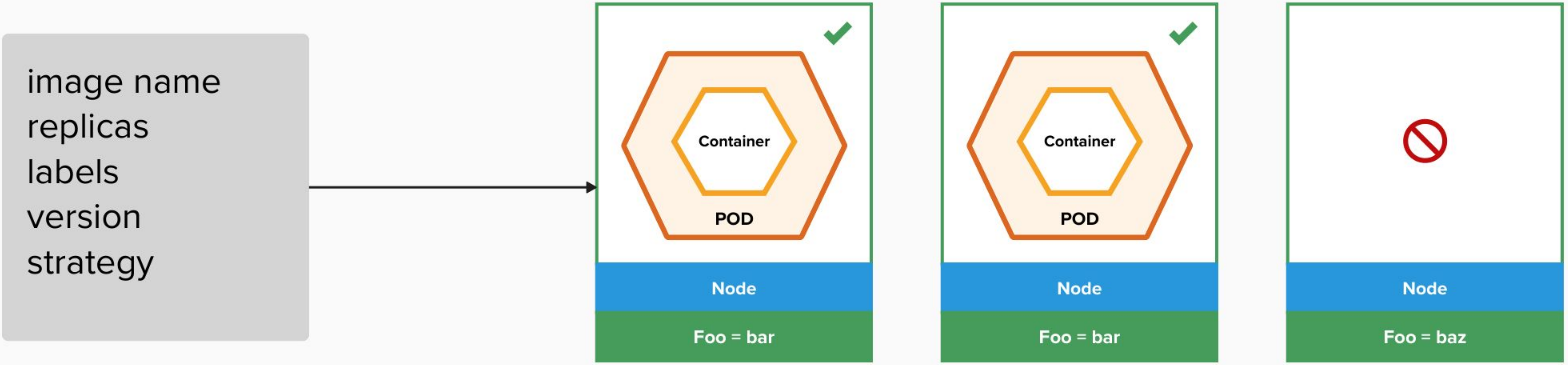
1. RECAP

Deployments and DeploymentConfigurations
define how to roll out new versions of Pods



1. RECAP

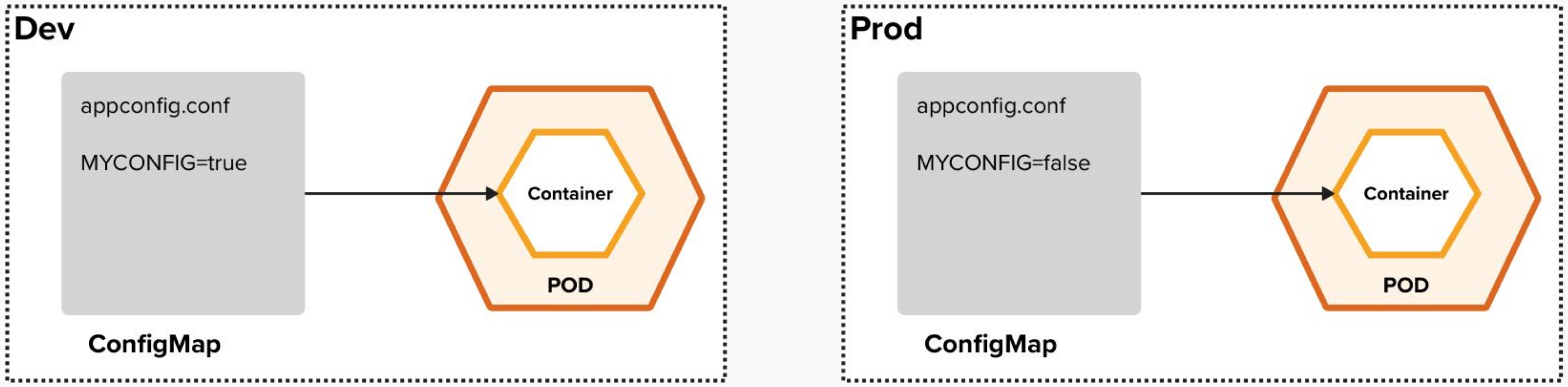
a daemonset ensures that all (or some) nodes run a copy of a pod



DeamonSet

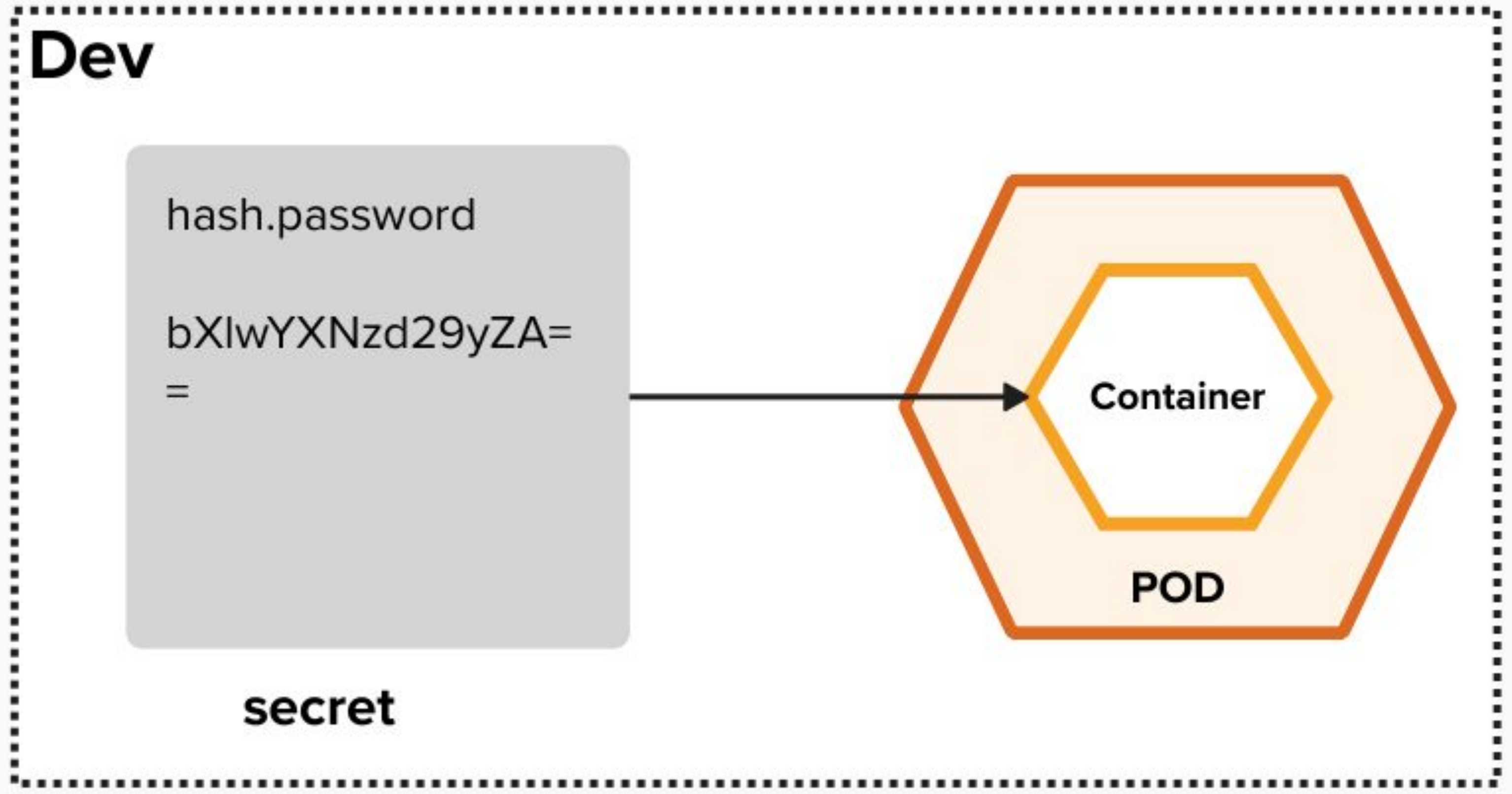
1. RECAP

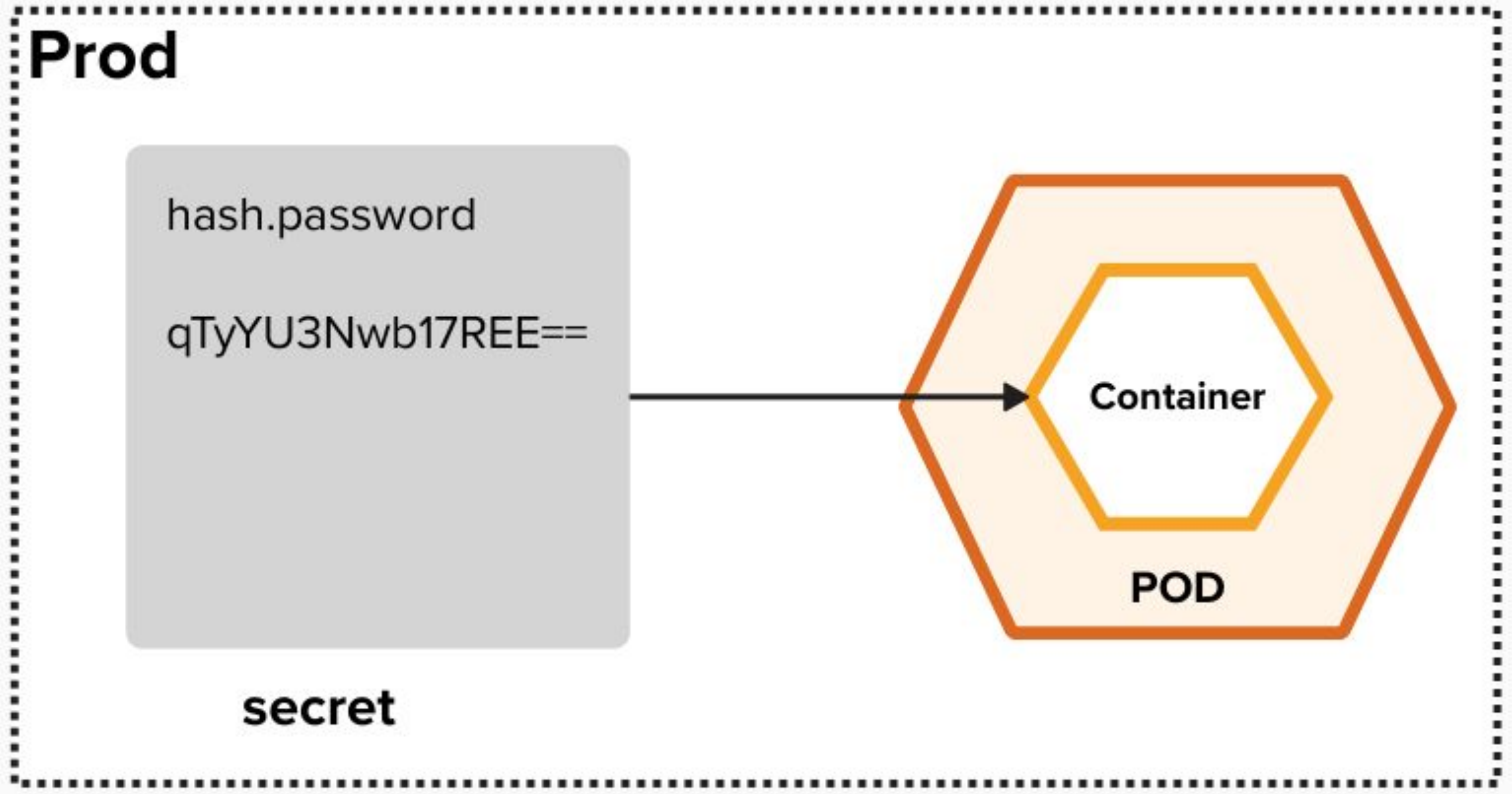
configmaps allow you to decouple configuration artifacts from image content



1. RECAP

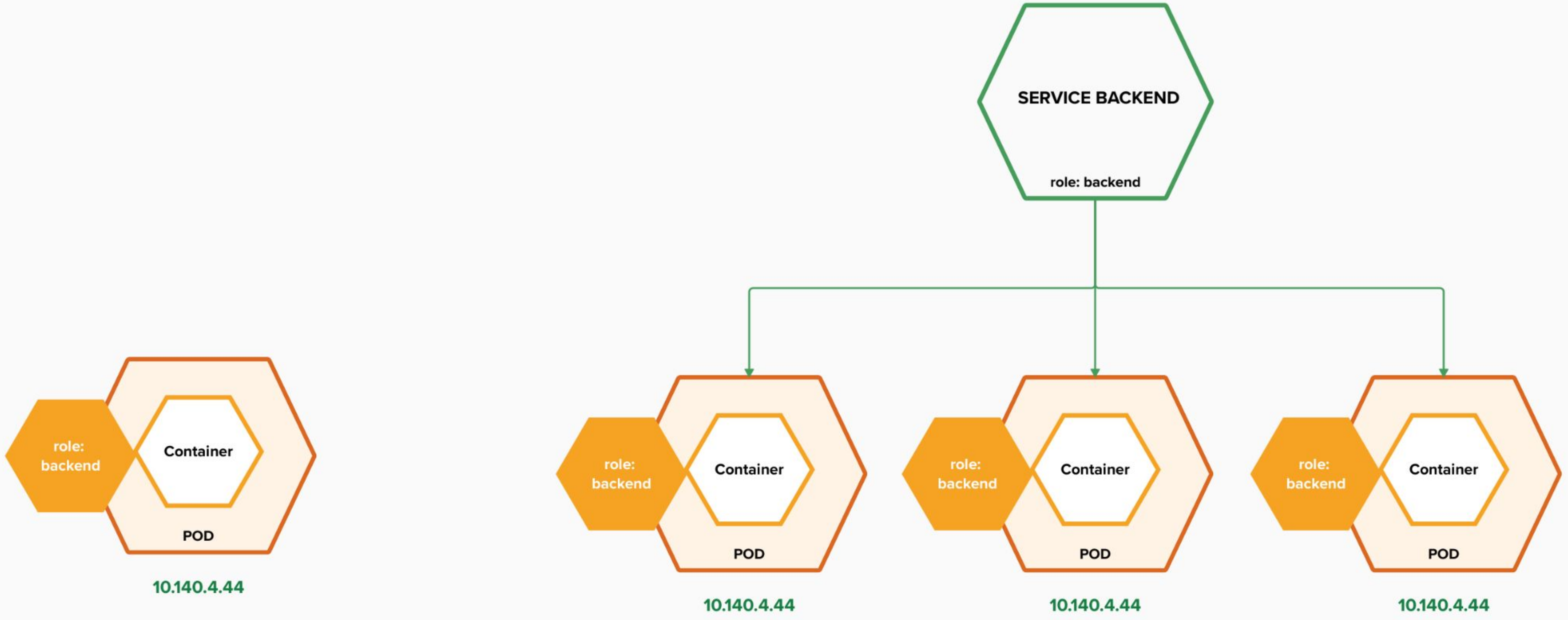
secrets provide a mechanism to hold sensitive information such as passwords





1. RECAP

services provide internal load-balancing and
service discovery across pods



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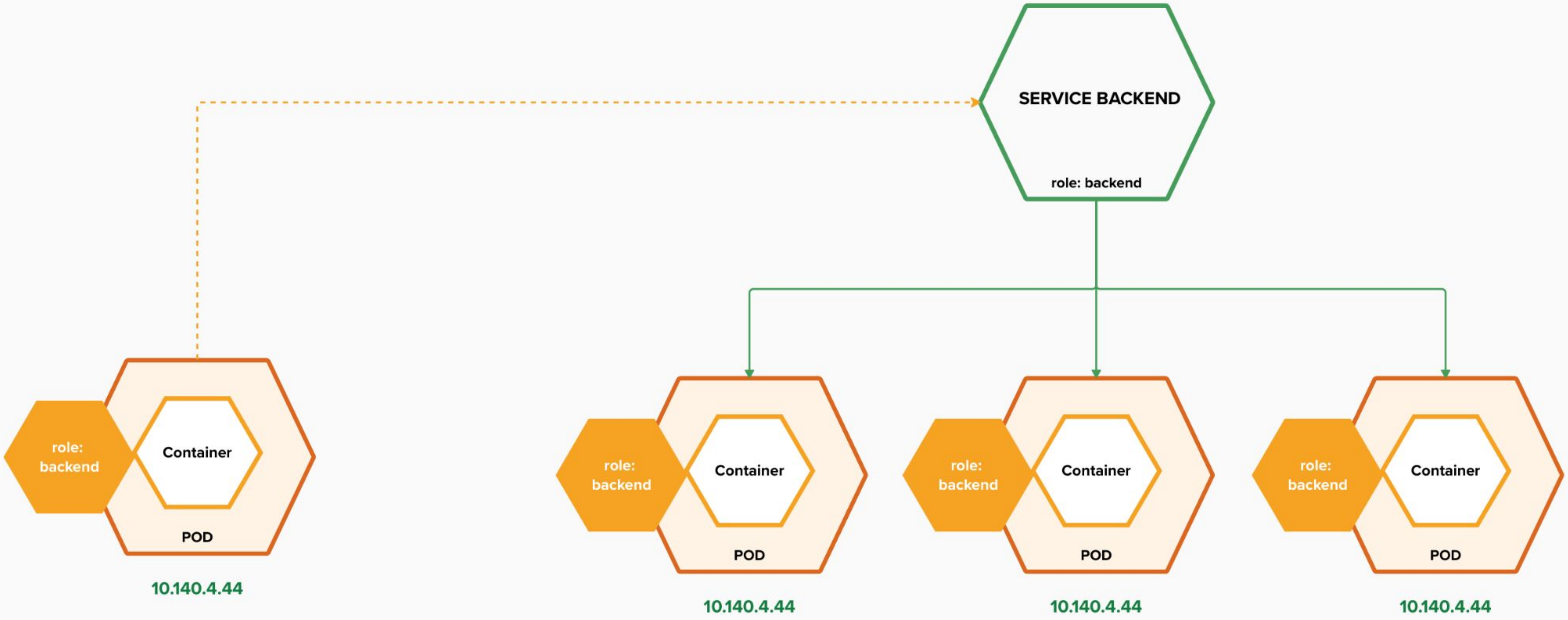
1. RECAP

apps can talk to each other via services



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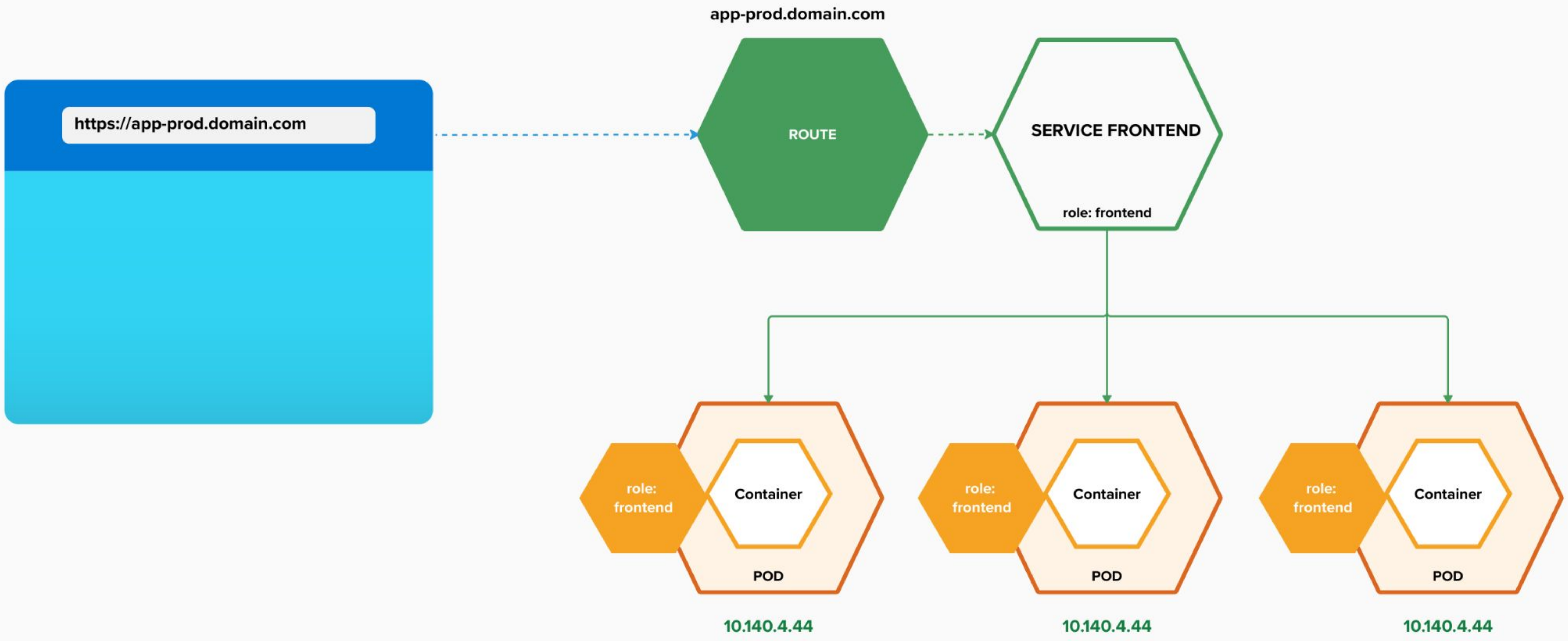
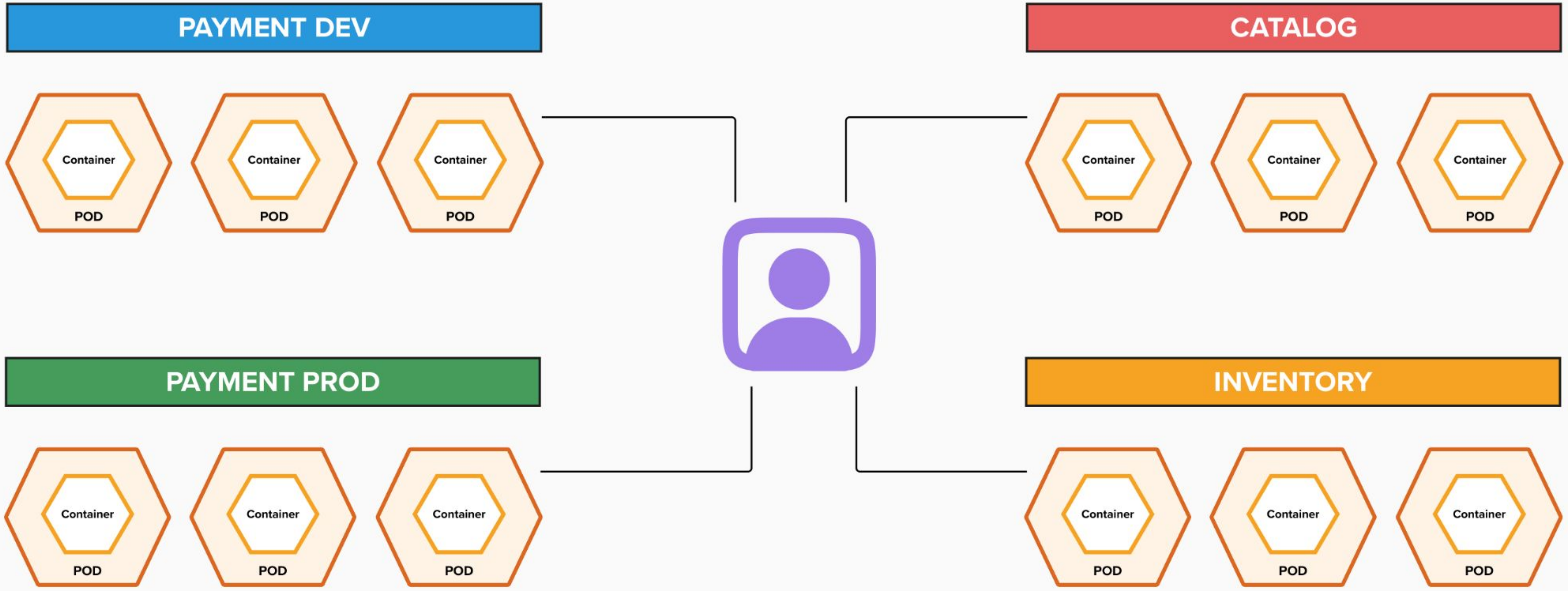
1. RECAP

routes make services accessible to clients outside the environment via real-world urls



1. RECAP

projects isolate apps across environments, teams, groups and departments



containers are created from container images

What is OpenShift?

Everything you need, out of the box

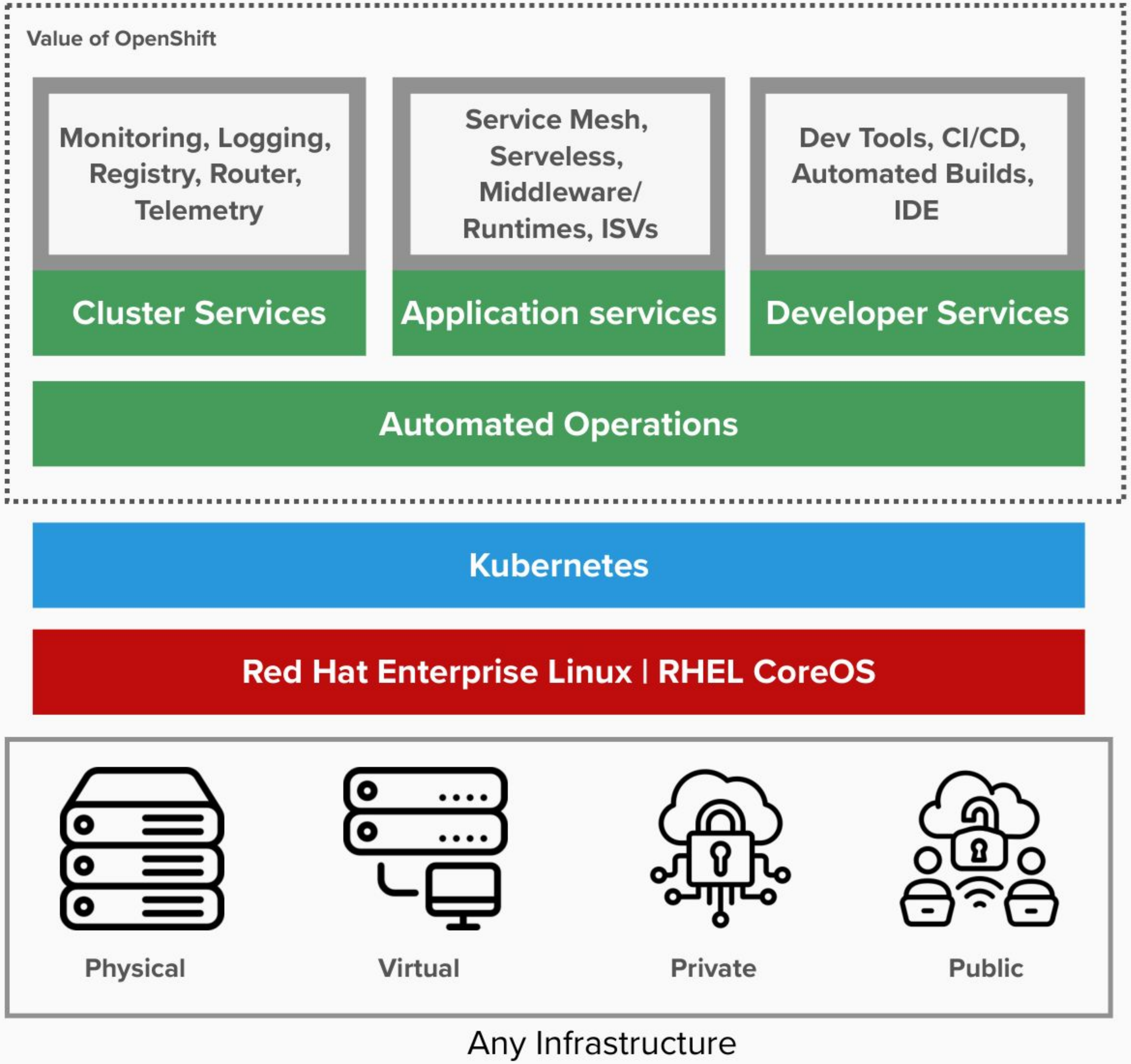
1. Fully integrated and automated architecture

2. Seamless Kubernetes deployment on any cloud or on-premises environment

3. Fully automated installation, from cloud infrastructure to OS to application services

4. One click platform and application updates

5. Auto-scaling of cloud resources



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a container is the smallest compute unit

OPENSHIFT ARCHITECTURE

2. OPENSHIFT ARCHITECTURE

your choice of infrastructure

2. OPENSHIFT ARCHITECTURE

workers run workloads

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2. OPENSHIFT ARCHITECTURE

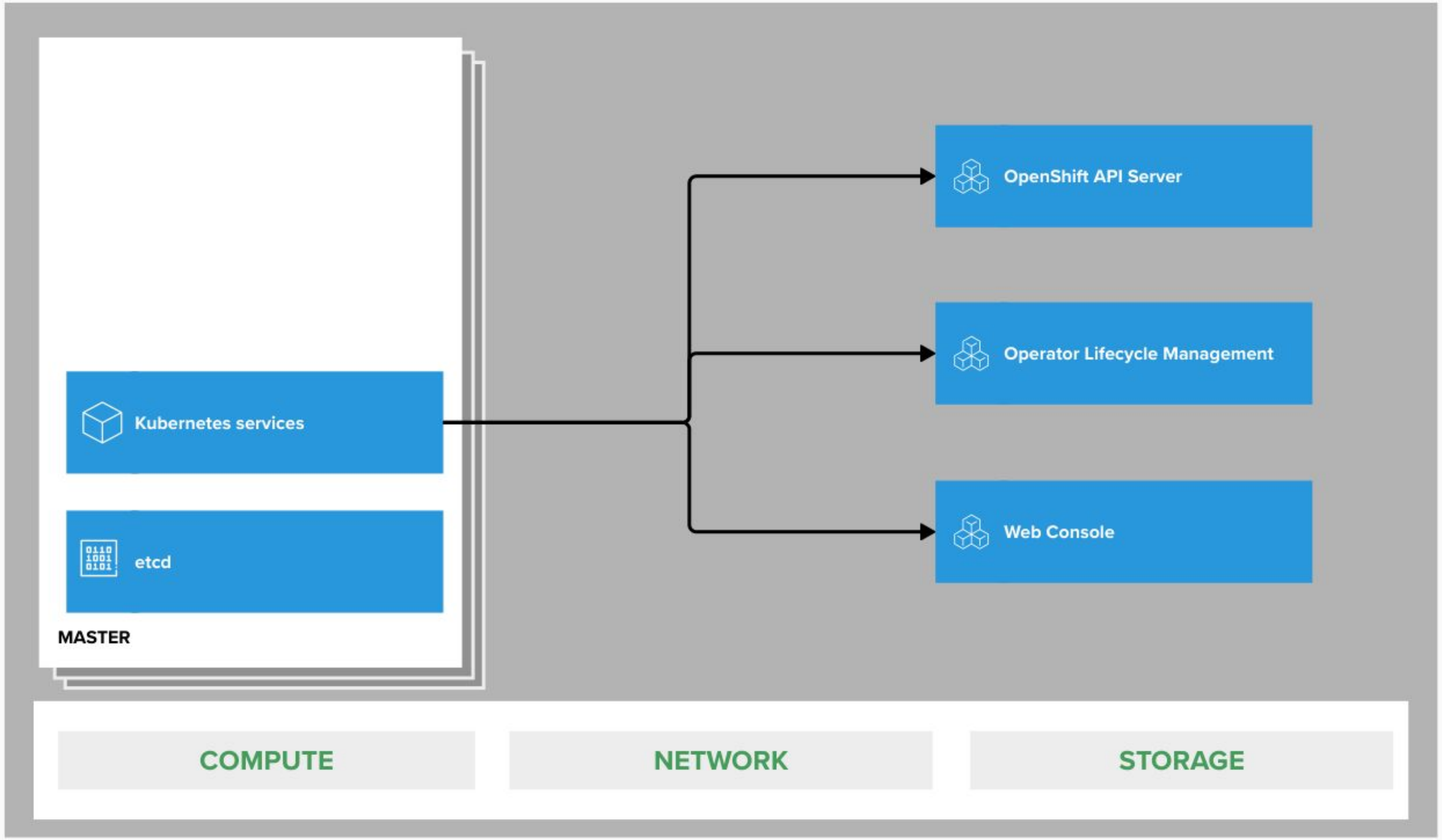
masters are the control plane

2. OPENSHIFT ARCHITECTURE

state of everything

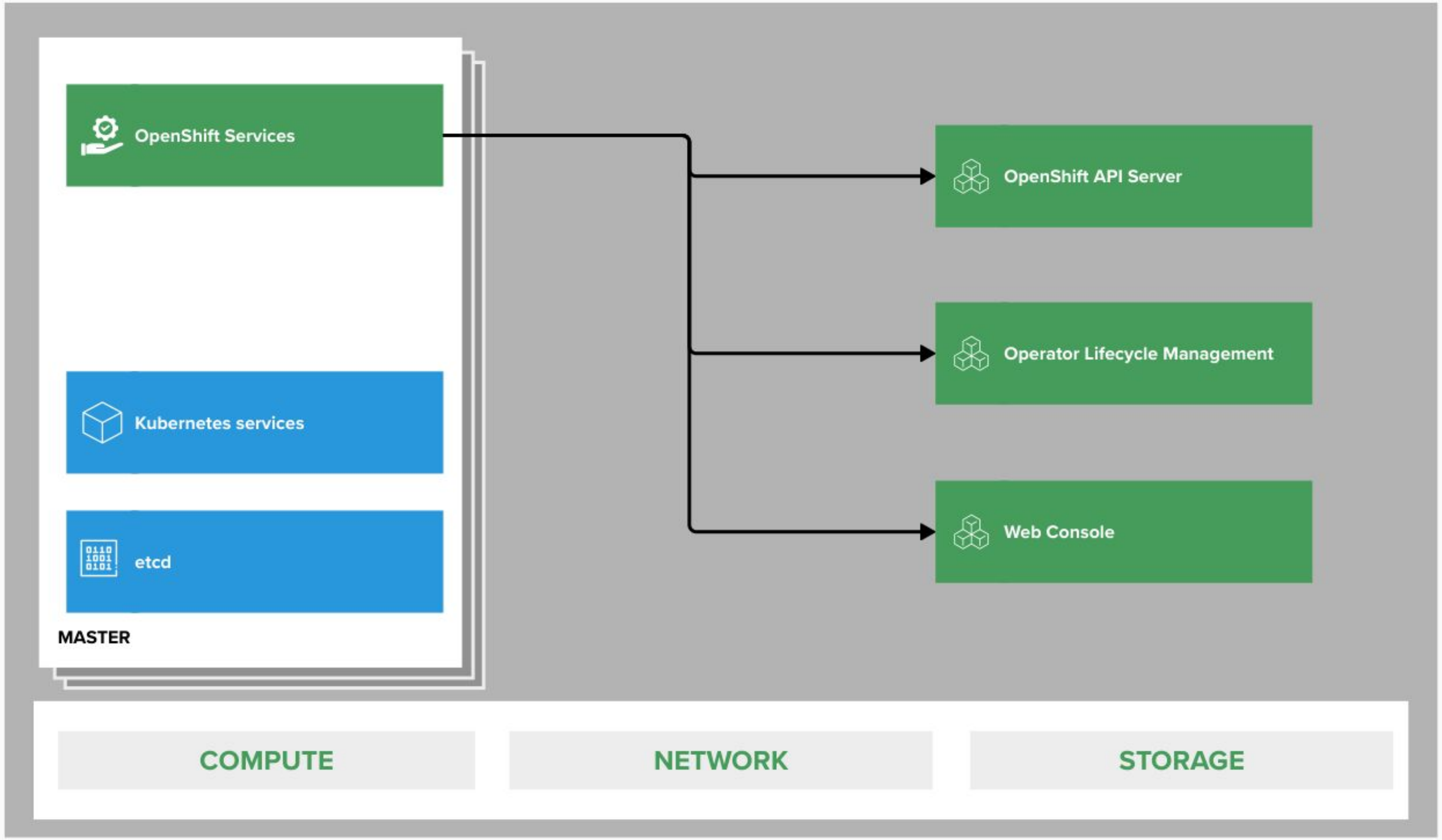
2. OPENSHIFT ARCHITECTURE

core kubernetes components



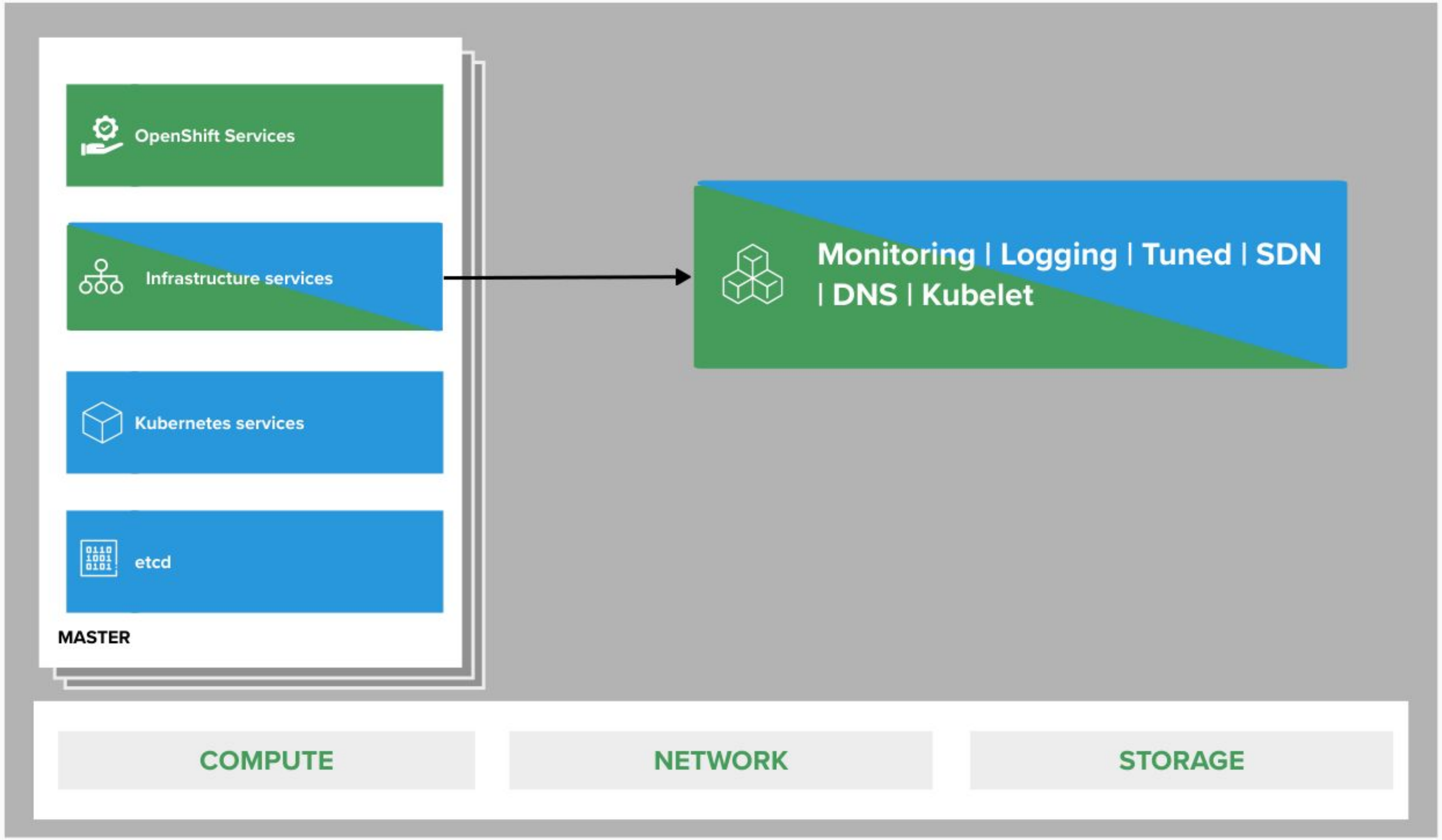
2. OPENSHIFT ARCHITECTURE

core OpenShift components



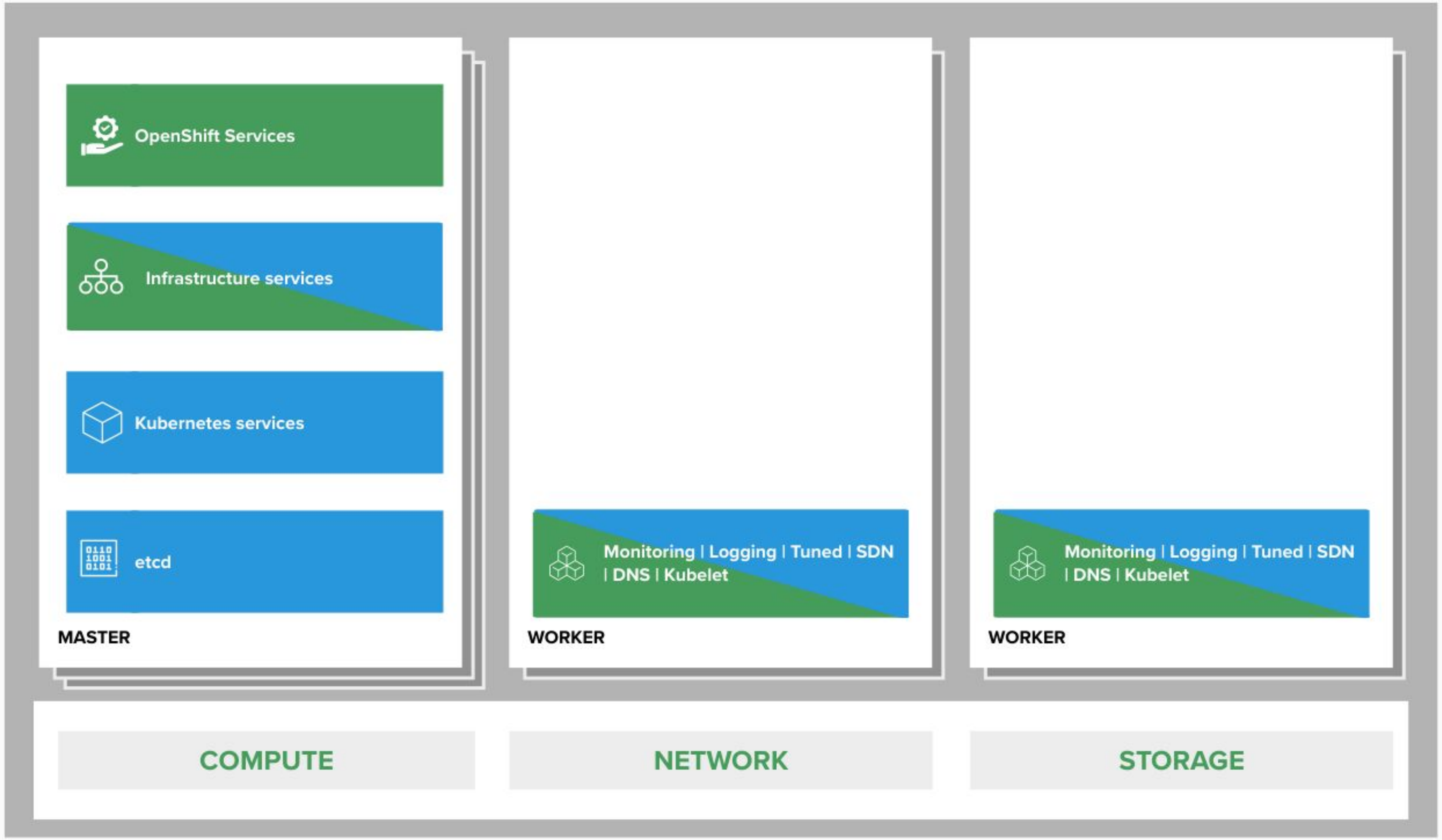
2. OPENSHIFT ARCHITECTURE

internal and support infrastructure services



2. OPENSHIFT ARCHITECTURE

run on all hosts



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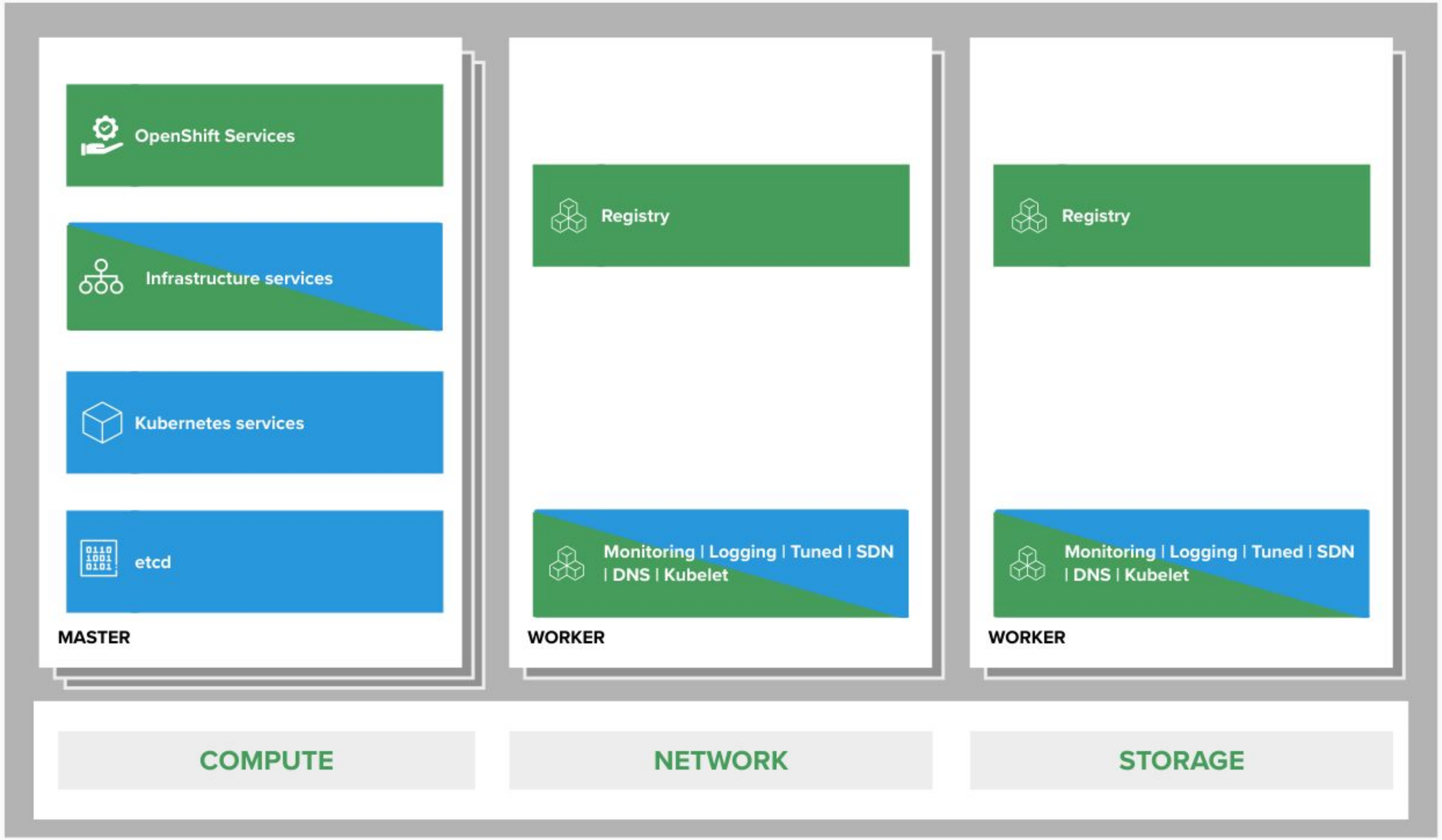
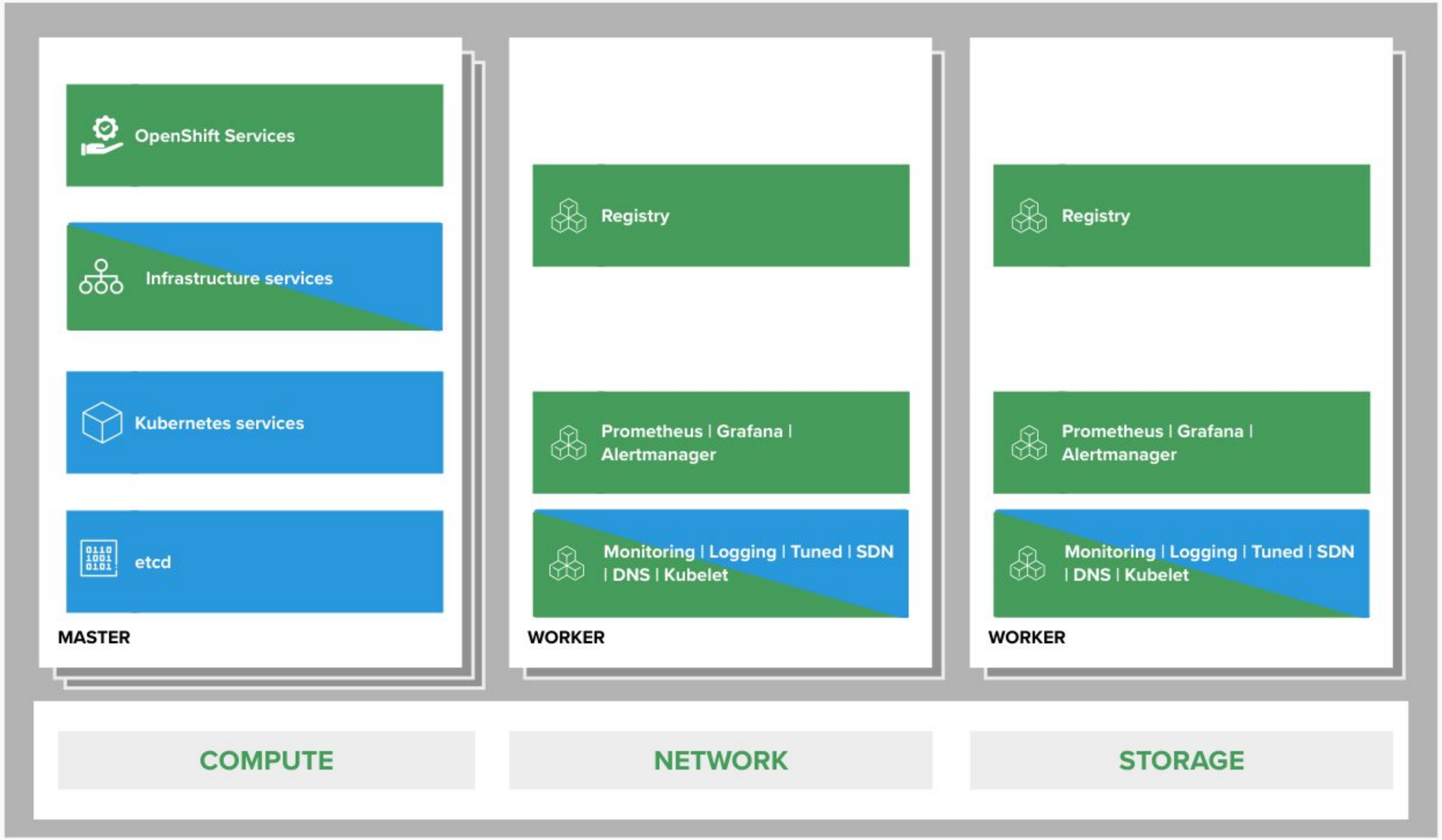
2. OPENSHIFT ARCHITECTURE

integrated image registry



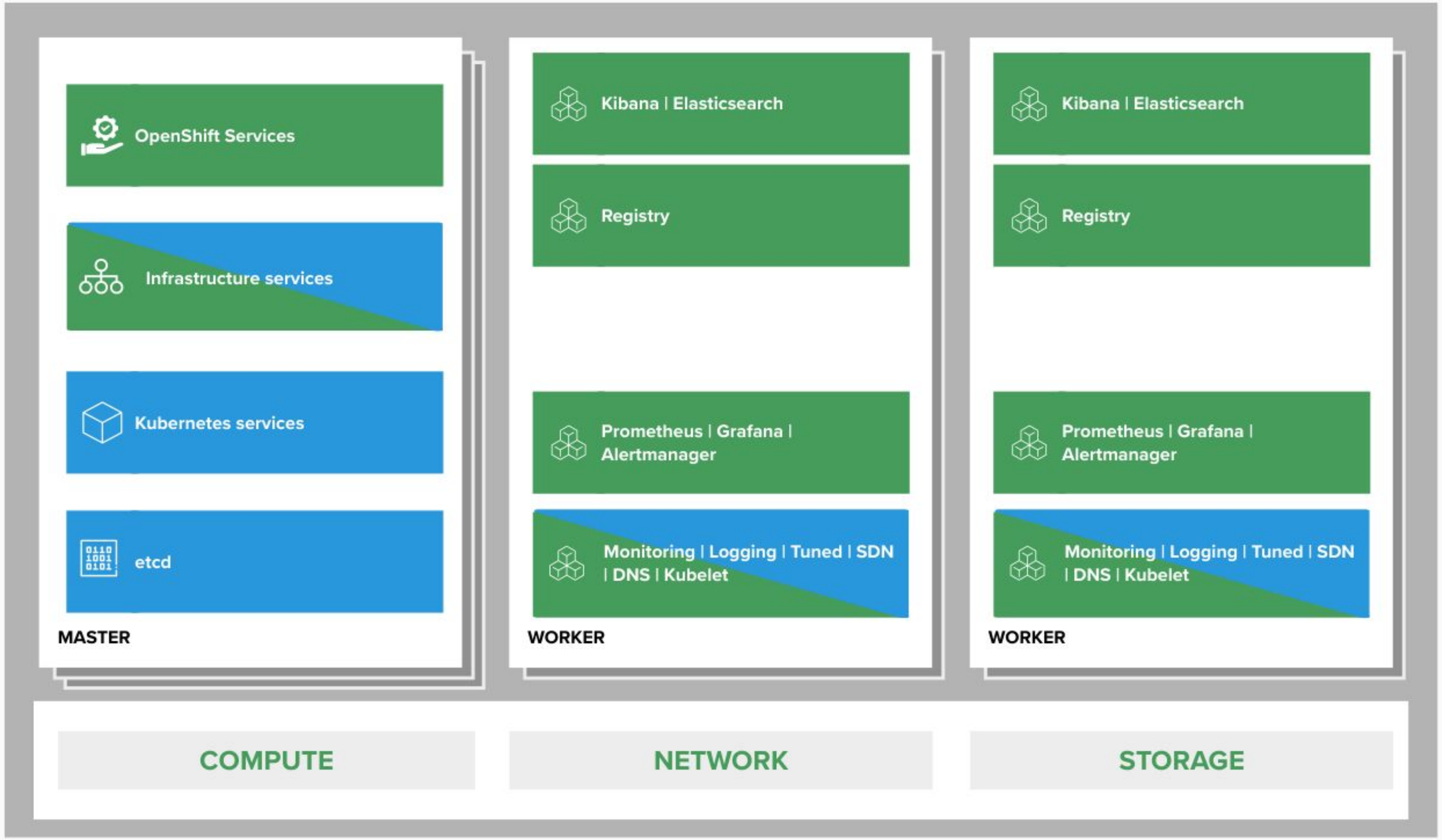
2. OPENSHIFT ARCHITECTURE

cluster monitoring



2. OPENSHIFT ARCHITECTURE

log aggregation



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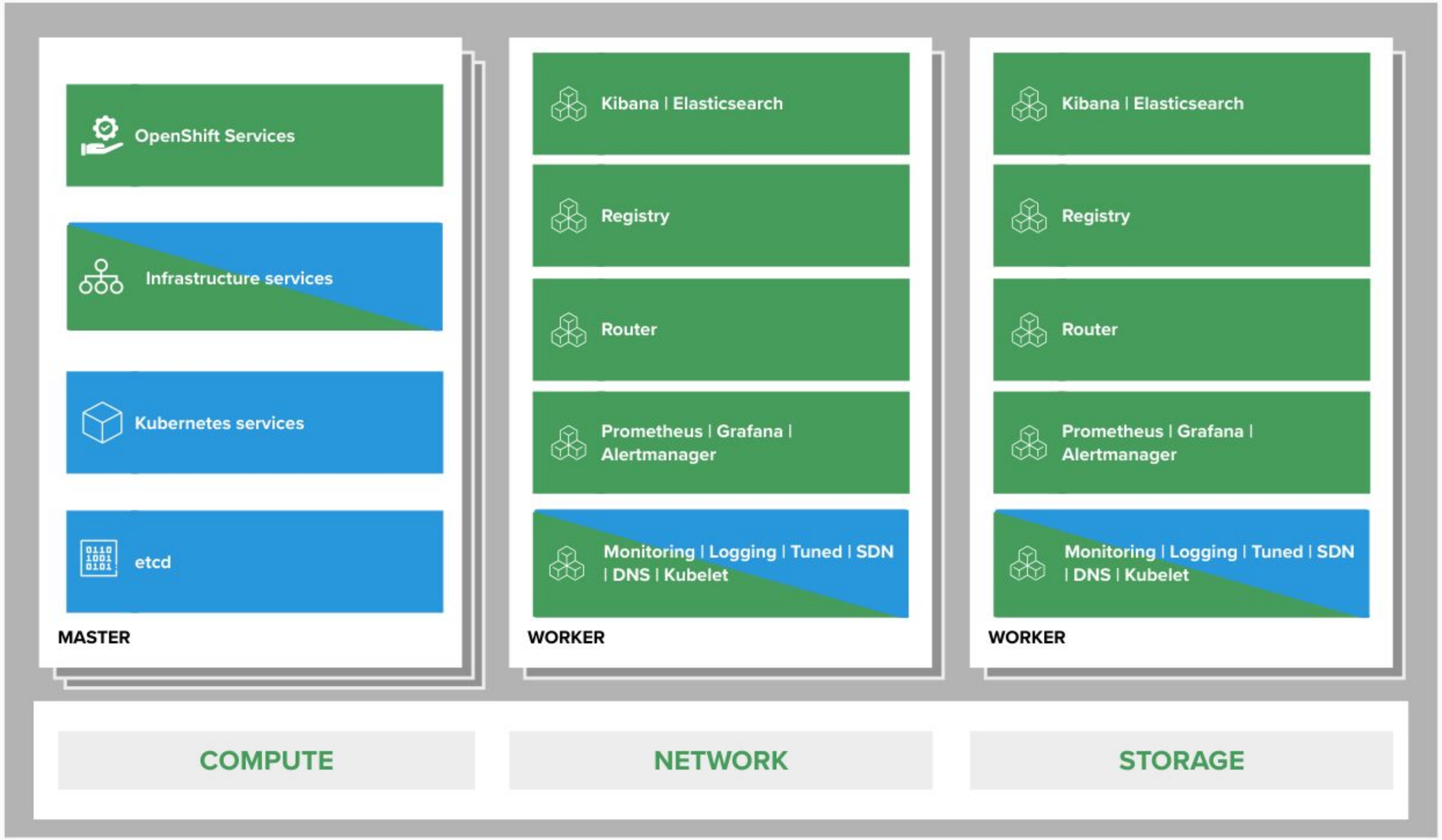
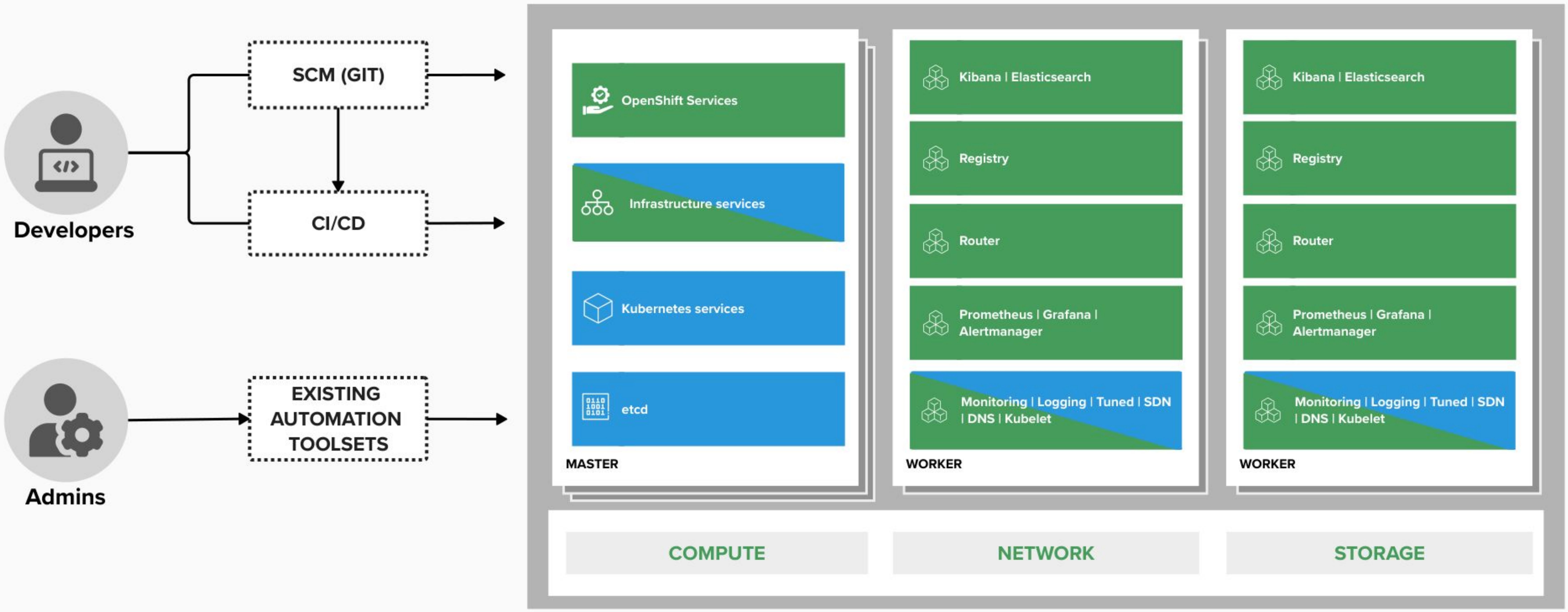
2. OPENSHIFT ARCHITECTURE

integrated routing



2. OPENSHIFT ARCHITECTURE

dev and ops via web, cli, API, and IDE



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containers are created from container images

OpenShift lifecycle, installation & upgrades

Red Hat Enterprise Linux CoreOS

	RED HAT® ENTERPRISE LINUX®	RED HAT® ENTERPRISE LINUX CoreOS
	General Purpose OS	**Immutable container host**
BENEFITS	• 10+ year enterprise life cycle • Industry standard security • High performance on any infrastructure • Customizable and compatible with wide ecosystem of partner solutions	• Self-managing, over-the-air updates • Immutable and tightly integrated with OpenShift • Host isolation is enforced via Containers • Optimized performance on popular infrastructure
WHEN TO USE	When customization and integration with additional solutions is required	When cloud-native, hands-free operations are a top priority

Immutable Operating System

Red Hat Enterprise Linux CoreOS is versioned with OpenShift

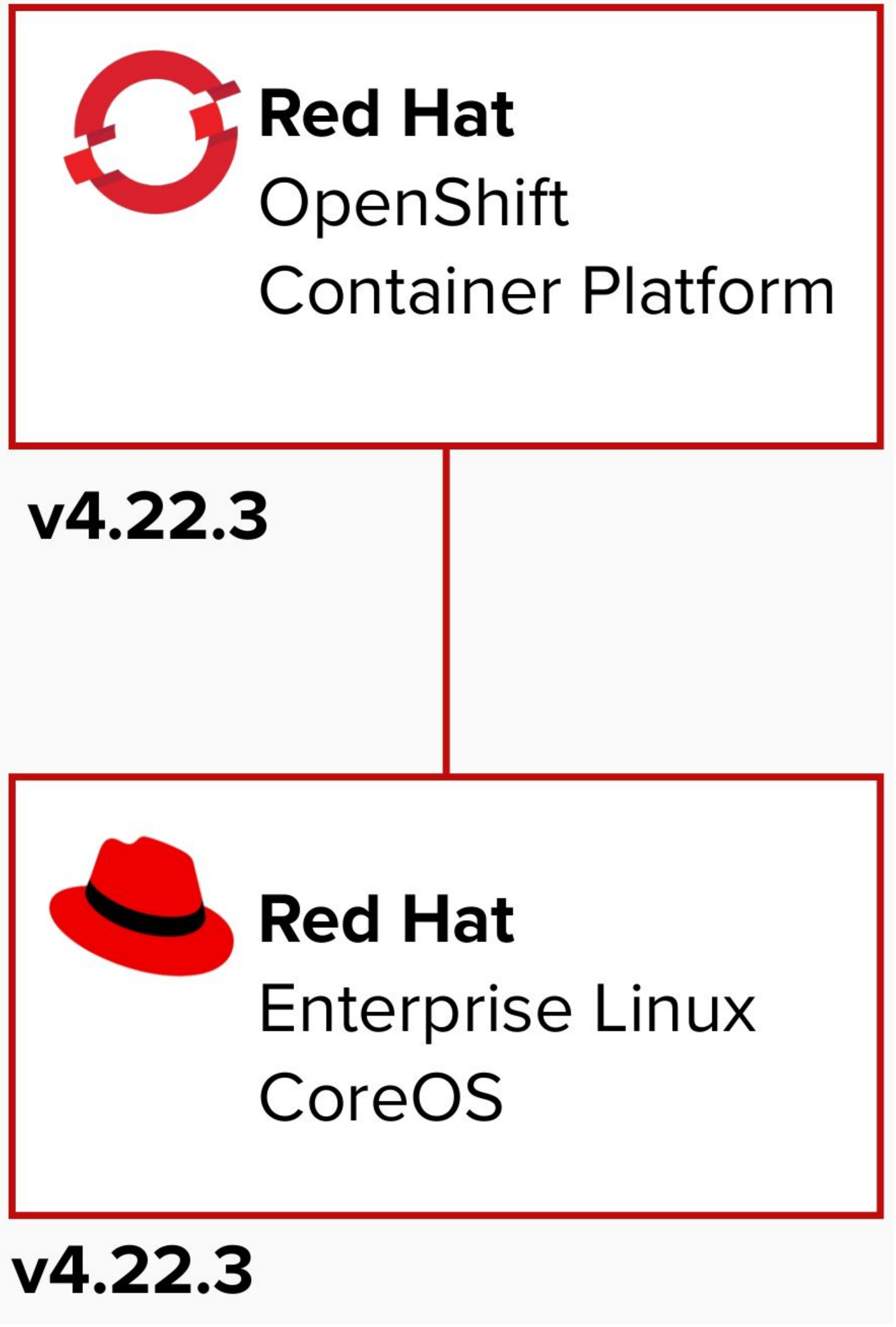
CoreOS is tested and shipped in conjunction with the platform. Red Hat runs thousands of tests against these configurations.

Red Hat Enterprise Linux CoreOS is managed by the cluster

The Operating system is operated as part of the cluster, with the config for components managed by Machine Config Operator:

- CRI-O config
- Kubelet config
- Authorized registries
- SSH config

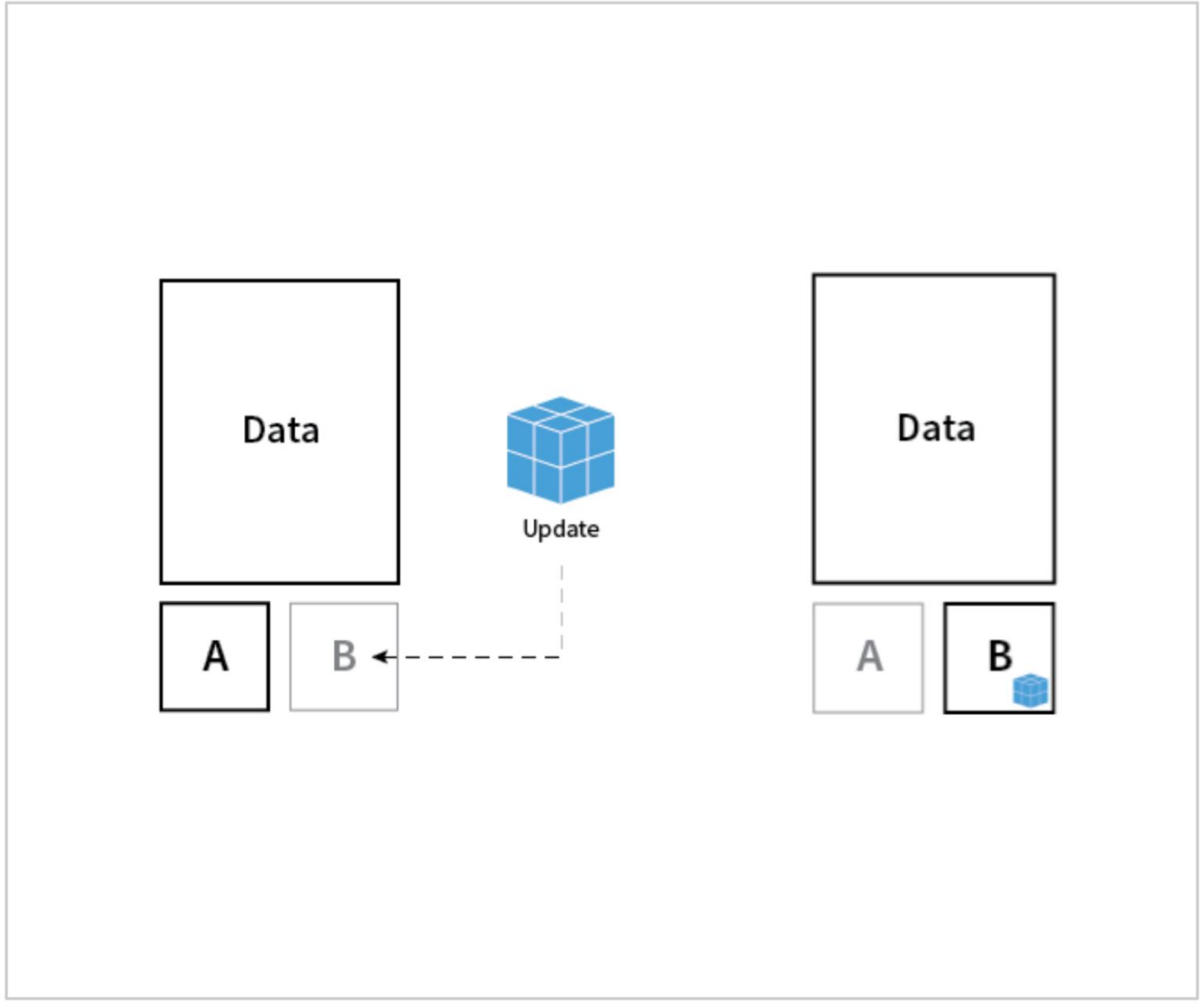
RHEL CoreOS admins are responsible for:
Nothing.



Transactional Updates via rpm-ostree

Transactional updates ensure that the Red Hat CoreOS is never altered during runtime. Rather it is booted directly into an always “known good” version.

- Each OS update is versioned and tested as an complete image.
- OS binaries (/usr) are read-only
- Updates encapsulated in container images
- file system and package layering available for hotfixes and debugging



Machine Config Operator (MCO)

Provides cluster-level configuration, enables rolling upgrades, and prevents drift between new and existing nodes. The MCO is the heart of what makes RHCOS a kube-native operating system.

Configure Kernel Arguments for the Cluster

- oc create -f 50-kargs.yaml
- oc edit mc/50-kargs

MCO can be paused to suspend operations

Provides control for when changes can be deployed

Custom MachinePools can have inheritance

Enables MachineConfigs to scale

```yaml
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 50-kargs
spec:
  KernelArguments:
    - audit=1
    - audit_backlog_limit=8192
    - net.ifnames.prefix=net
```

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CRI-O Support in OpenShift

CRI-O tracks and versions identical to Kubernetes, simplifying support permutations

CRI-O 1.34	Kubernetes 1.34	OpenShift 4.21
CRI-O 1.35	Kubernetes 1.35	OpenShift 4.22
CRI-O 1.36	Kubernetes 1.36	OpenShift 5.0

Installation

OPENSHIFT CONTAINER PLATFORM

HOSTED OPENSHIFT

Full Stack Automated

Simplified opinionated "Best Practices" for cluster provisioning

Fully automated installation and updates including host container OS.

Red Hat Enterprise Linux CoreOS

IPI (Installer-Provisioned Infrastructure)

Pre-existing Infrastructure

Customer managed resources & infrastructure provisioning

Plug into existing DNS and security boundaries

Red Hat Enterprise Linux CoreOS

Red Hat Enterprise Linux

UPI (User-Provisioned Infrastructure)

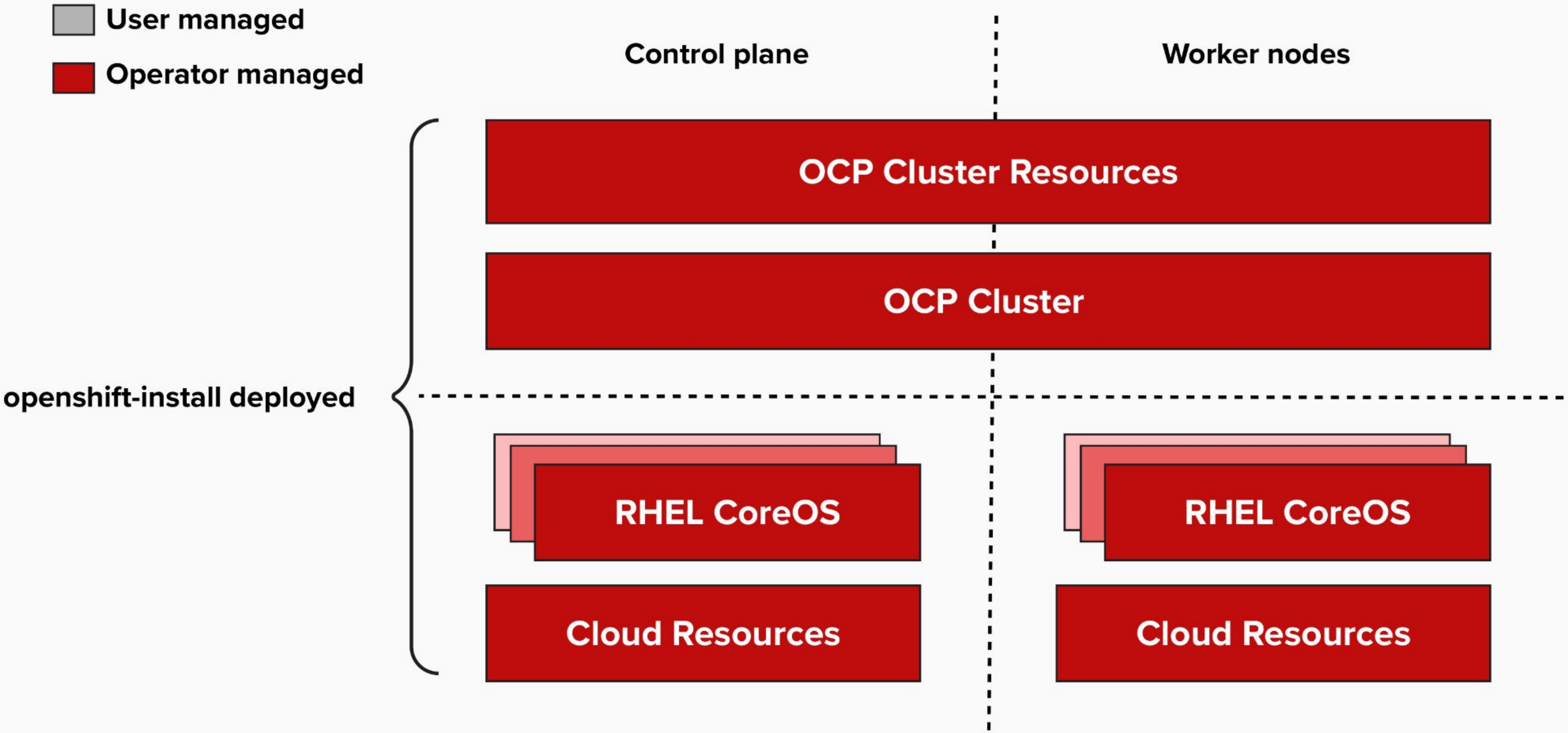
Cloud Provider

Deploy directly from a cloud provider. Jointly managed by Red Hat engineers and cloud provider engineer .

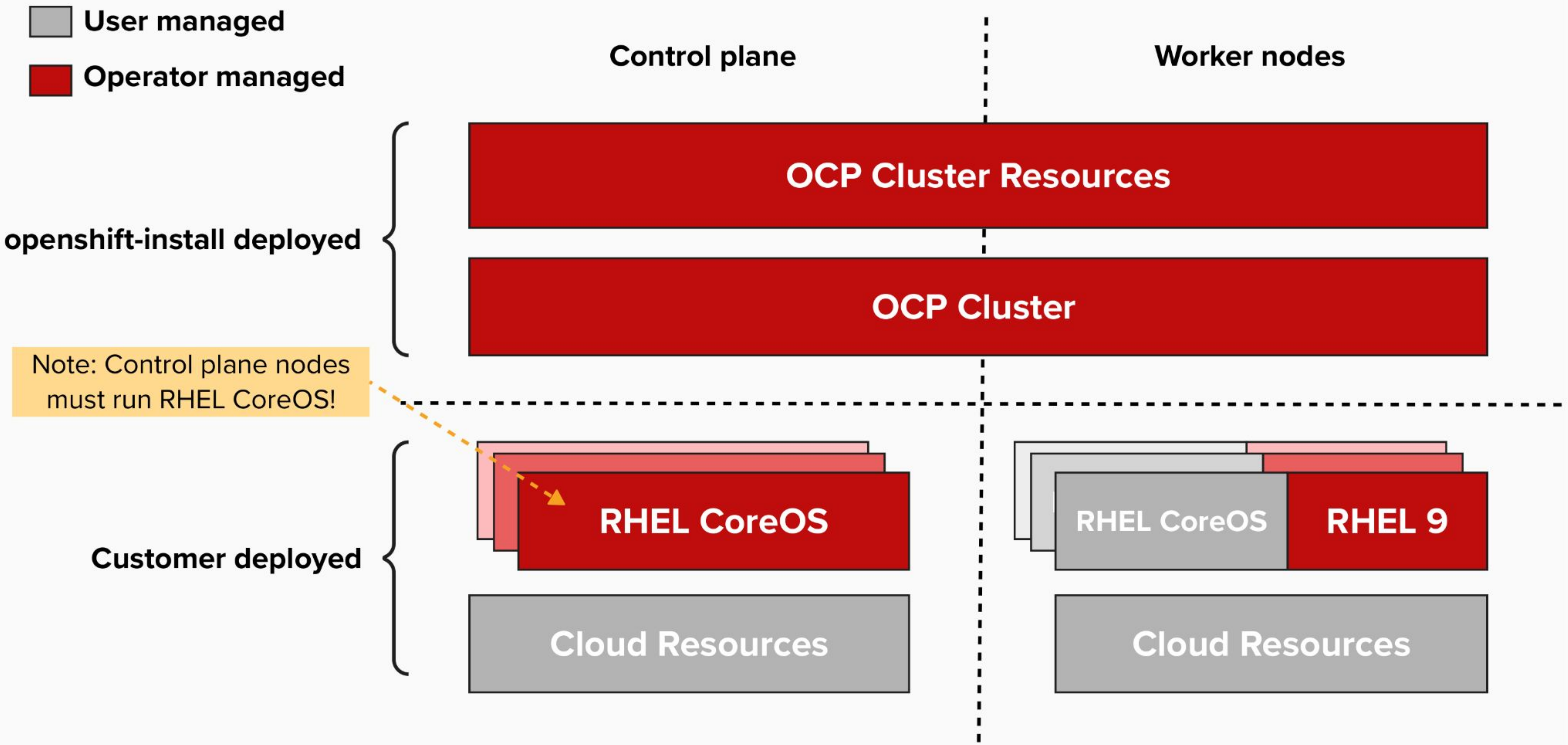
OpenShift Dedicated

Get a powerful cluster, fully Managed by Red Hat engineers and support.

Full-stack Automated Installation



Pre-existing Infrastructure Installation



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Comparison of Paradigms

	Full Stack Automation	Pre-existing Infrastructure
Build Network	Installer	User
Setup Load Balancers	Installer	User
Configure DNS	Installer	User
Hardware/VM Provisioning	Installer	User
OS Installation	Installer	User
Generate Ignition Configs	Installer	Installer
OS Support	Installer: RHEL CoreOS	User: RHEL CoreOS + RHEL 9
Node Provisioning / Autoscaling	yes	Only for providers with OpenShift Machine API support

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Supported Providers

Full Stack Automation (IPI)

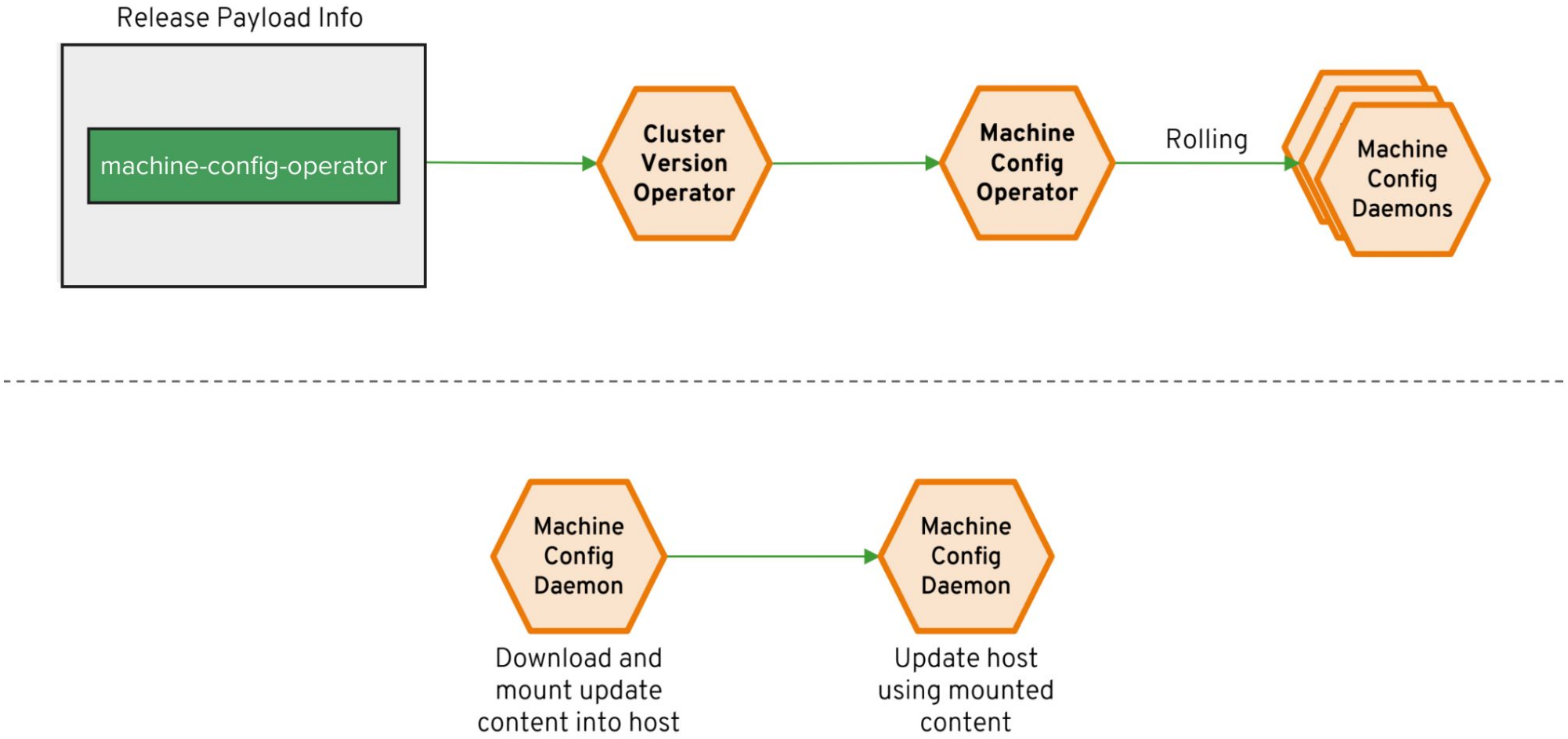
- aws
- Azure
- Google Cloud
- Azure Stack
- openstack.
- IBM Cloud
- vmware aws
- Alibaba Cloud

Pre-existing Infrastructure (UPI)

- aws
- Bare Metal
- Google Cloud
- IBM
  - IBM Z
  - IBM LinuxONE

Cluster Management

Over-the-air updates



Cluster Management

Rolling Machine Updates

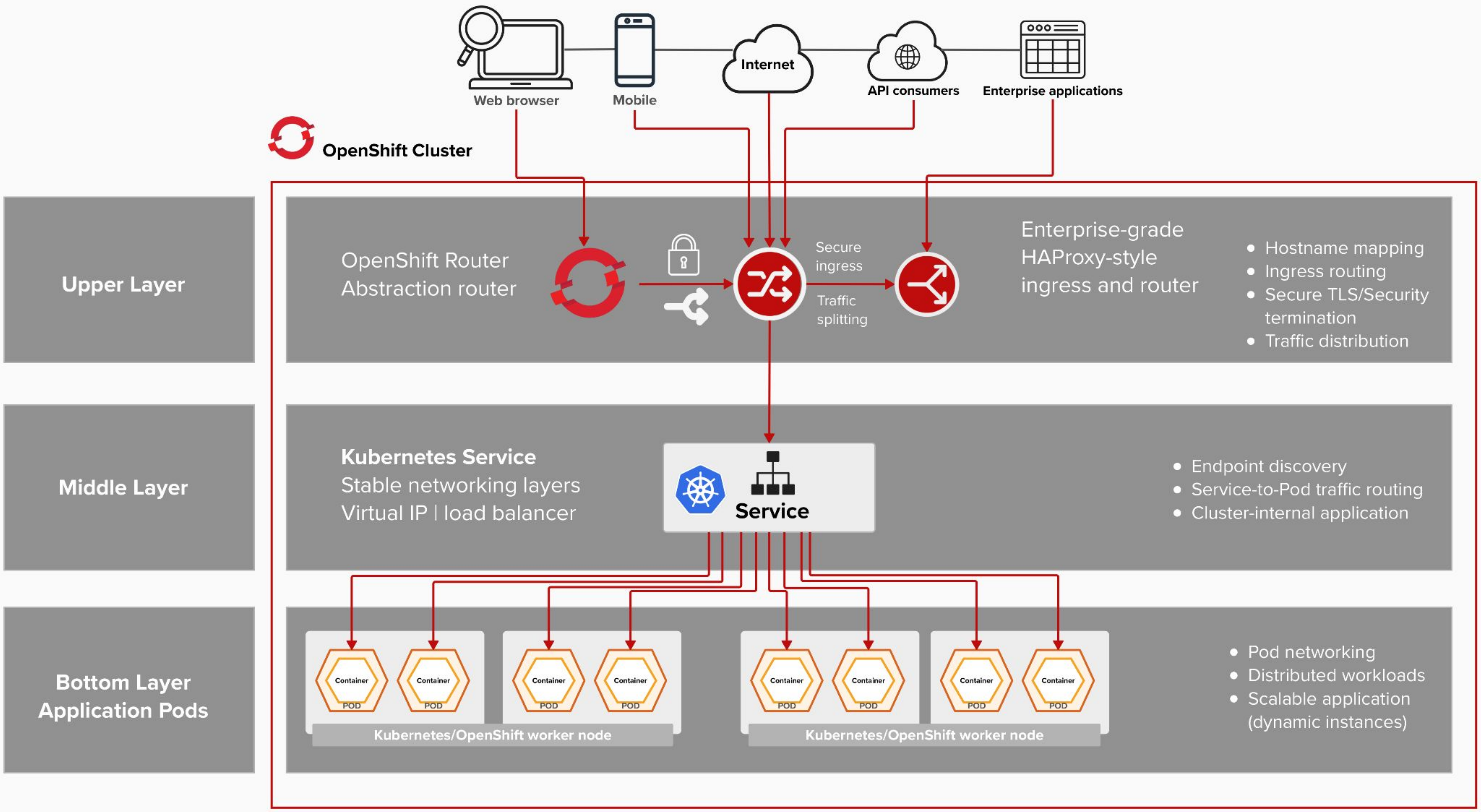
Single-click updates
- RHEL CoreOS version & config
- Kubernetes core components
- OpenShift cluster components

Machine Config Operator (MCO) controls updates
This is a DaemonSet that runs on all Nodes in the cluster. When you upgrade with oc adm upgrade, the MCO executes these changes.

containers are created from container images

Networking

Getting Traffic Into the Cluster



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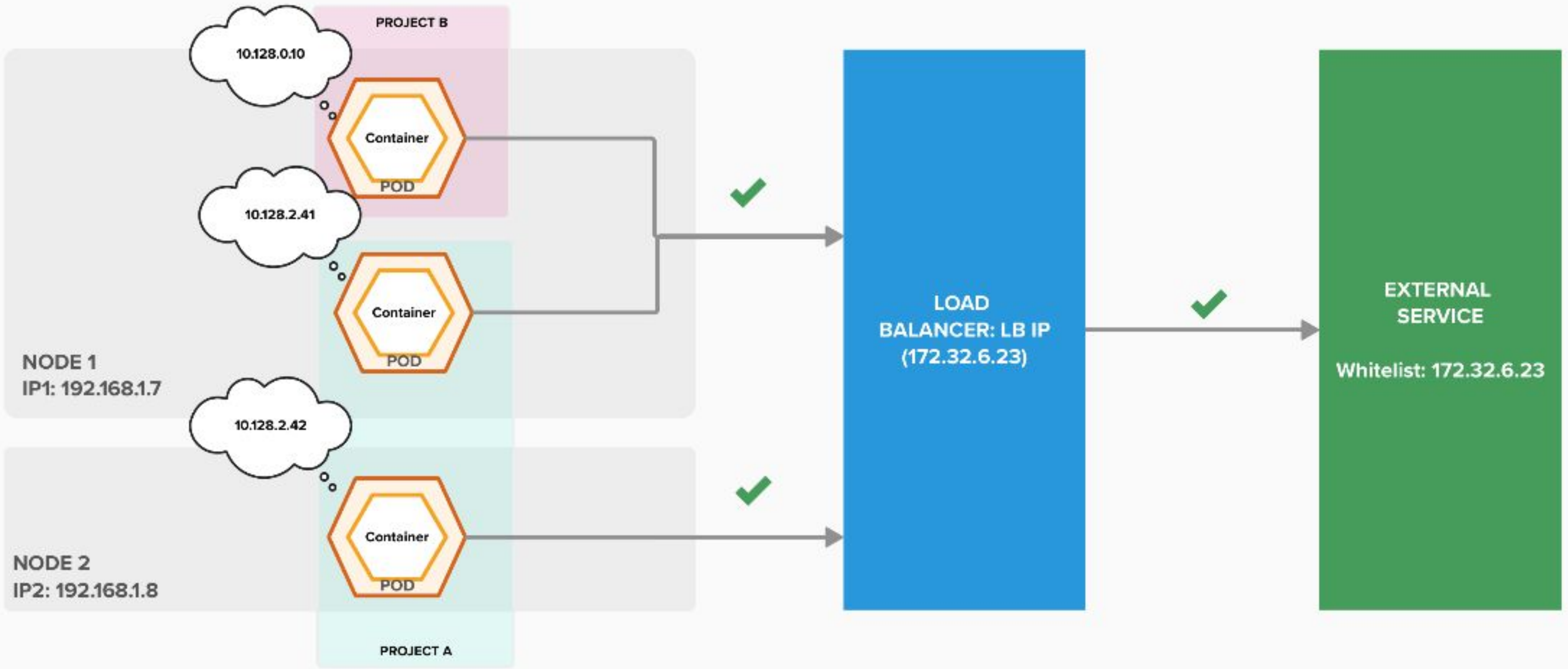
Getting Traffic Out the Cluster

SNAT

10.128.0.10 -> 192.168.1.7
10.128.2.41 -> 192.168.1.7

10.128.2.42 -> 192.168.1.8



SNAT **LB**

10.128.0.10 -> 192.168.1.7 -> 172.32.6.23
10.128.2.41 -> 192.168.1.7 -> 172.32.6.23

10.128.2.42 -> 192.168.1.8 -> 172.32.6.23

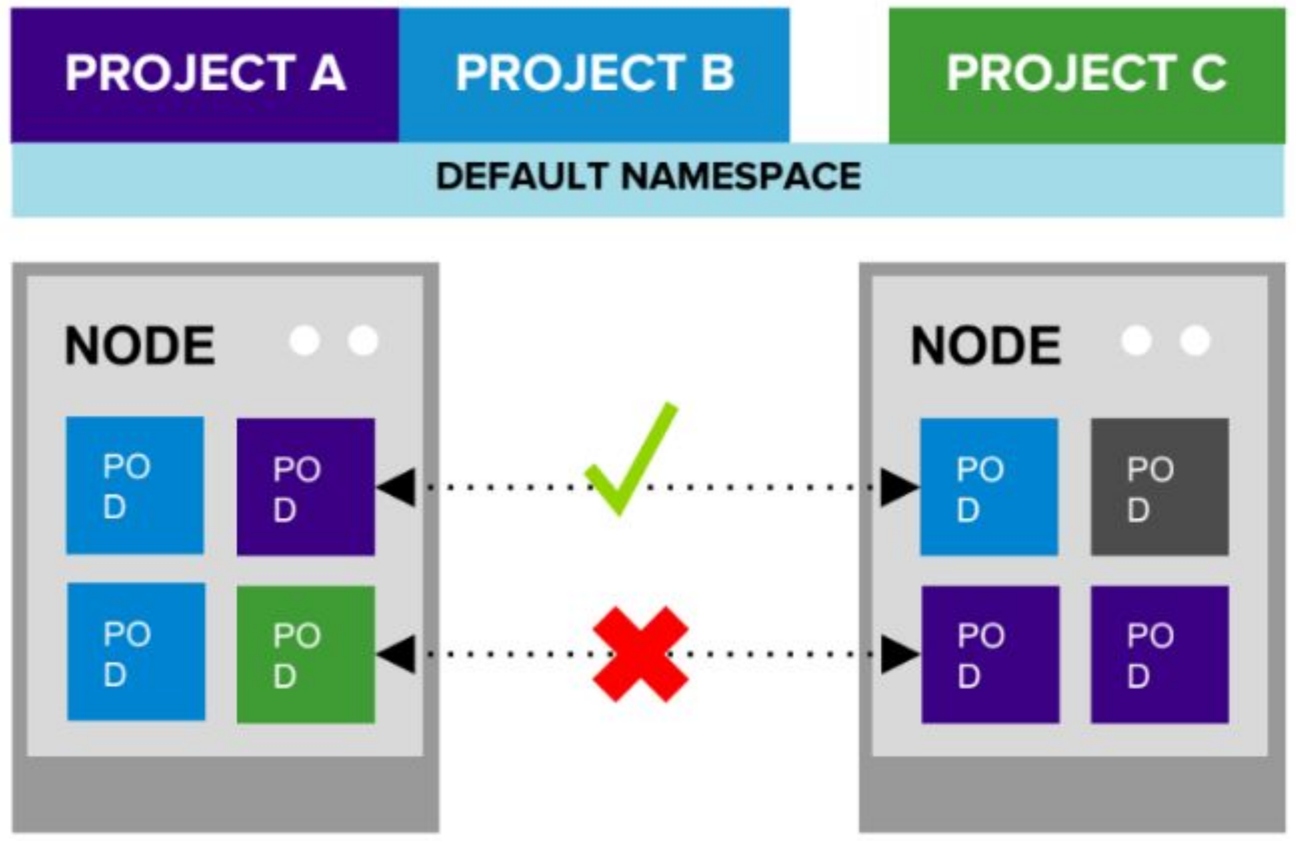
Getting Traffic Around the Cluster

OpenShift Network Plugins

Subnet
All pods can communicate with all other pods

Multitenant
Project level isolation.

Network Policy (Default)
Granular policy based isolation



Multitenant Network

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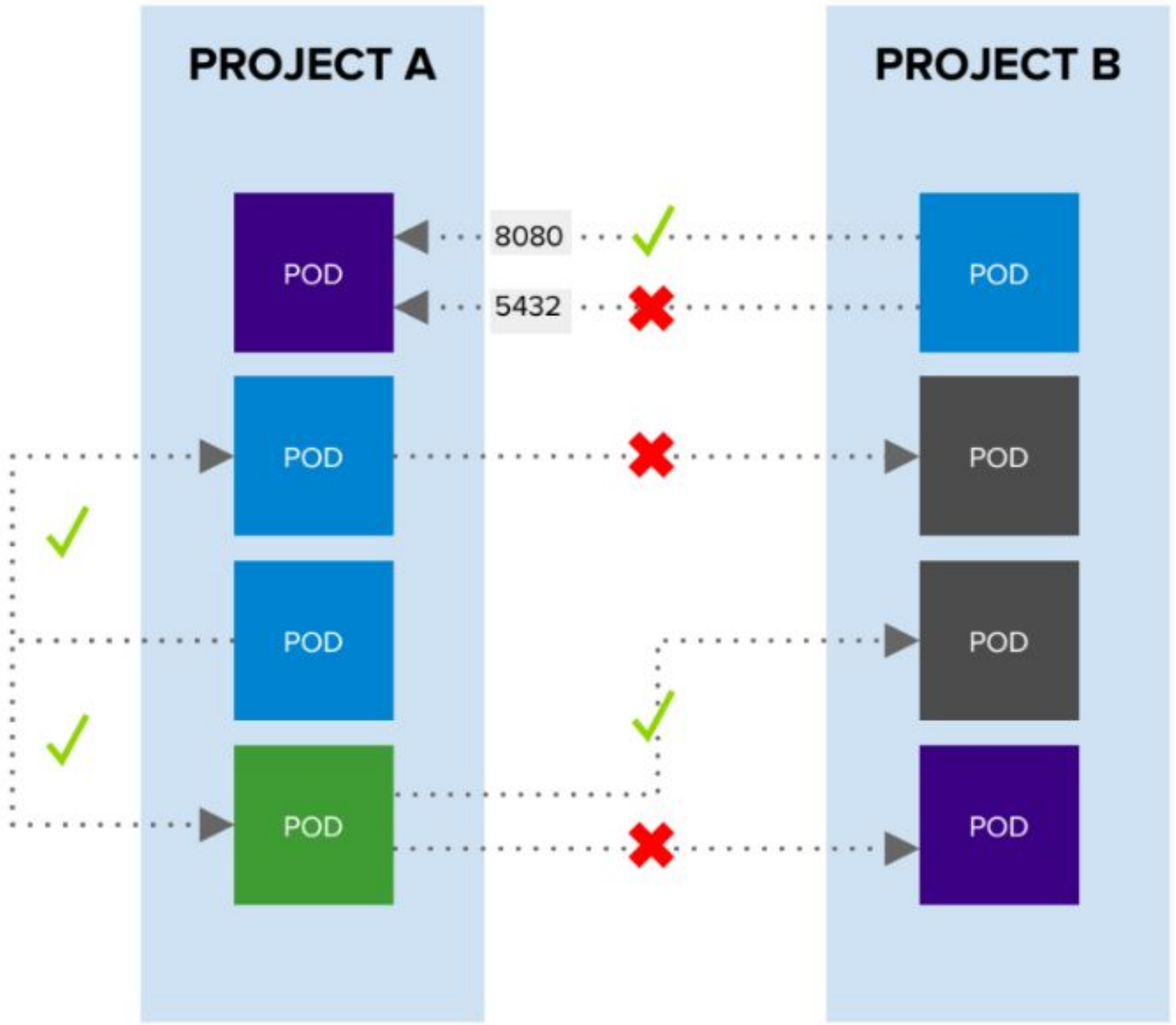
Getting Traffic Around the Cluster

Network Policy



Example Policies

Allow all traffic inside the project
Allow traffic from green to gray
Allow traffic to purple on 8080

```
apiVersion: extensions/v1beta1
kind: NetworkPolicy
metadata:
  name: allow-to-purple-on-8080
spec:
  podSelector:
    matchLabels:
      color: purple
  ingress:
  - ports:
    - protocol: tcp
      port: 8080
```

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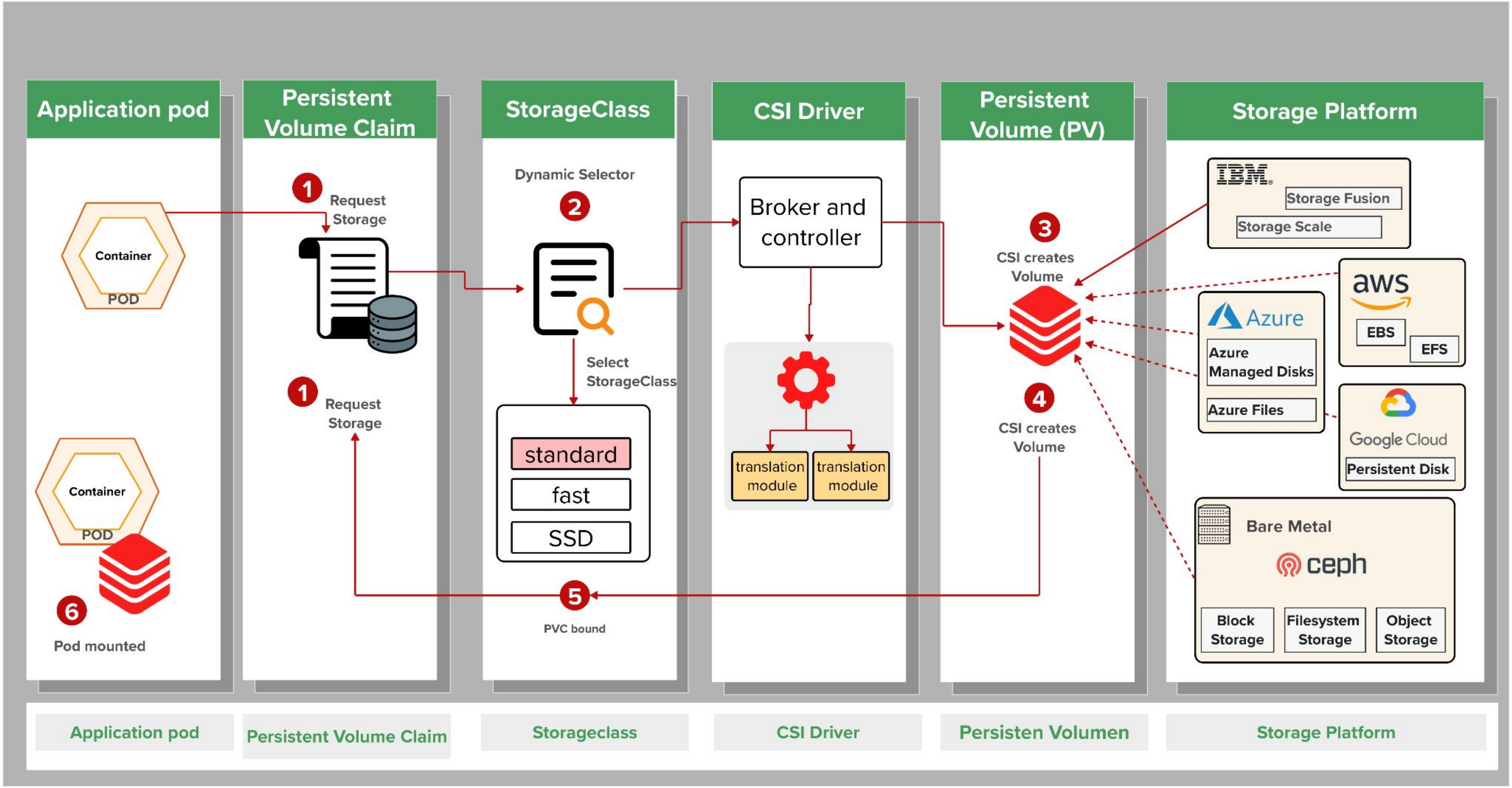
containers are created from container images

Persistent Storage



Application pod
Persistent Volume Claim
StorageClass
CSI Driver
Persistent Volume (PV)
Storage Platform
Container
POD
1
Request Storage
Dynamic Selector
2
Select StorageClass
standard
fast
SSD
Broker and controller
translation module
translation module
3
CSI creates Volume
4
CSI creates Volume
IBM
Storage Fusion
Storage Scale
aws
EBS
EFS
Azure
Azure Managed Disks
Azure Files
Google Cloud
Persistent Disk
Bare Metal
ceph
Block Storage
Filesystem Storage
Object Storage
1
Request Storage
Container
POD
6
Pod mounted
5
PVC bound
Application pod
Persistent Volume Claim
Storageclass
CSI Driver
Persisten Volumen
Storage Platform

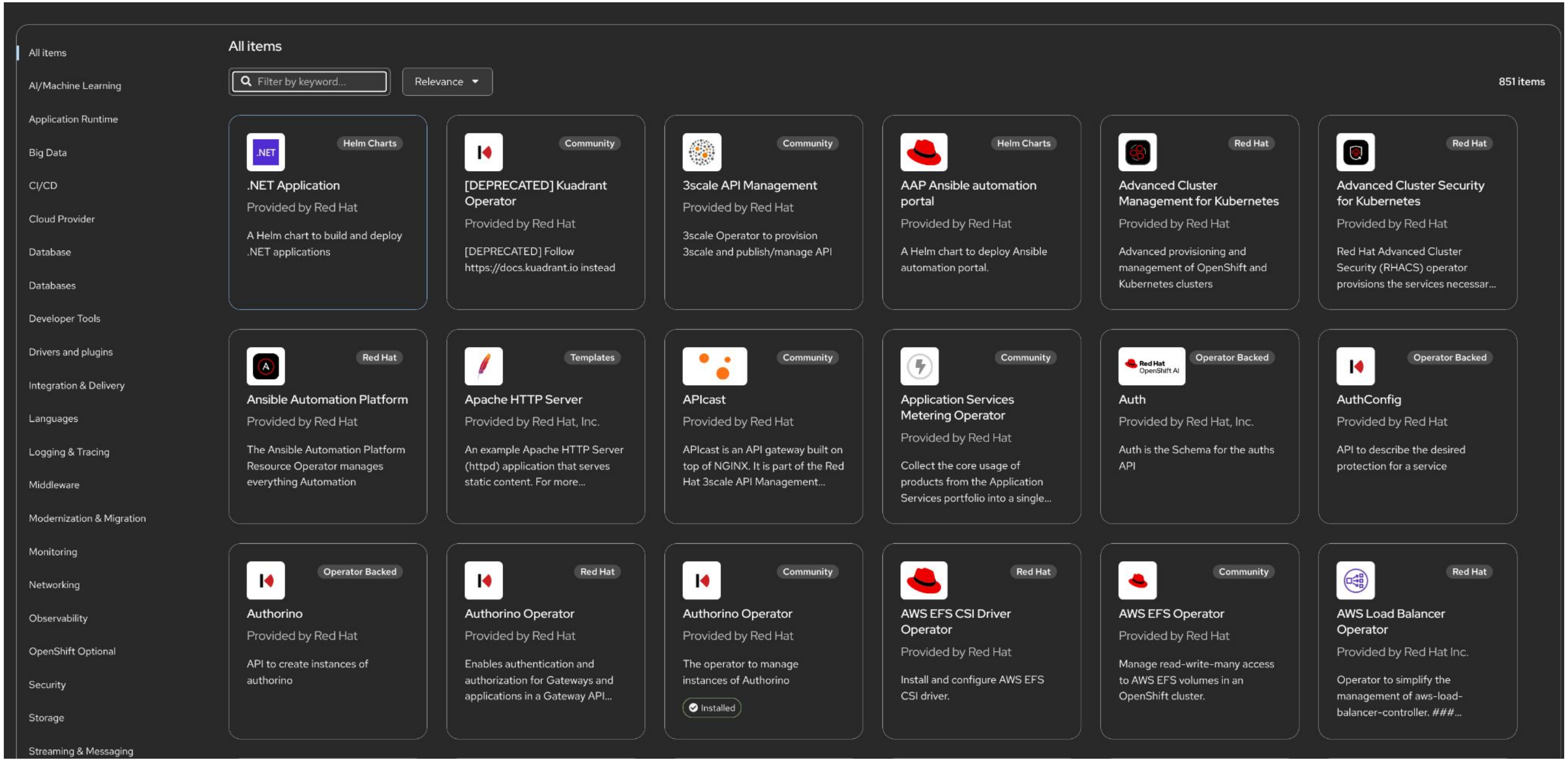
containers are created from container images

Operators

Red Hat Certified Operators

DEVOPS	dynatrace, Kong, joget, OpsMx, CITRIX
APM	APPDYNAMICS, INSTANA, New Relic., Sysdig, turbonomic, sematext
DATA SERVICES	GIGASPACES, hazelcast, PlanetScale, PERCONA, PROPHETSTOR, Starburst
DATABASE	MEMSQL, Couchbase, mongoDB., NUODB, PingCAP, crunchy data, MariaDB Foundation
SECURITY	aqua, anchore, BLACKDUCK BY SYNOPSYS, TREMOLO SECURITY, tufin, Twistlock
STORAGE	ROBIN, STORAGEOS, portworx

Operator Catalog



Operators as a First-Class Citizen

YourOperator v1.1.2
Bundle

OPERATOR
LIFECYCLE MANAGER

Deployment
Role
ClusterRole
RoleBinding
ClusterRoleBinding
ServiceAccount
CustomResourceDefinition

Operator Deployment
Custom Resource
Definitions
RBAC
API Dependencies
Update Path
Metadata

Operator Lifecycle Management